# Analog Electronics

**Classroom Workbook**

**ELECTRONICS AND COMM. ENGINEERING**

**ELECTRICAL ENGINEERING**

**INSTRUMENTATION ENGINEERING**

**GATE/ESE**



# Analog Electronics

## Classroom Workbook

**For EC, EE and EI**

**Copyrights © All Rights Reserved**

**Sorting HAT Technologies Pvt. Ltd.**

No part of this publication may be reproduced or distributed in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise or stored in a database or retrieval system without the prior written permission of the publishers. The program listings (if any) may be entered, stored and executed in a computer system, but they may not be reproduced for publication.

Printing of books passes through many stages - writing, composing, proof reading, printing etc. We try our level best to make the book error- free. If any mistake has inadvertently crept in, we regret it and would be deeply indebted to those who point it out. We do not take any legal responsibility.

**Edition : AE-2024/2025**

**Sorting HAT Technologies Pvt. Ltd.**

353, 12th'A'cross, off double road, Opp. City Centre

Residency, Indira nagar, 1st stage Bangalore

Bangalore KA 560038 IN.

# GATE Syllabus

**ECE**

Diode circuits: clipping, clamping and rectifiers. BJT and MOSFET amplifiers: biasing, ac coupling, small signal analysis, frequency response. Current mirrors and differential amplifiers. Op-amp circuits: Amplifiers, summers, differentiators, integrators, active filters, Schmitt triggers and oscillators.

**EE**

Simple diode circuits: clipping, clamping, rectifiers; Amplifiers: biasing, equivalent circuit and frequency response; oscillators and feedback amplifiers; operational amplifiers: characteristics and applications; single stage active filters, Active Filters: Sallen Key, Butterwoth, VCOs and timers

**IN**

Characteristics and applications of diode, Zener diode, BJT and MOSFET; small signal analysis of transistor circuits, feedback amplifiers. Characteristics of ideal and practical operational amplifiers; applications of opamps: adder, subtractor, integrator, differentiator, difference amplifier, instrumentation amplifier, precision rectifier, active filters, oscillators, signal generators, voltage controlled oscillators and phase locked loop, sources and effects of noise and interference in electronic circuits.

# ESE Syllabus

**ET**

Small signal equivalent circuits of diodes, BJTS and FETs; Diode circuits for different uses; Biasing and stability of BJT and JFET amplifier circuits; Analysis and design of amplifiers, single or multi-stage; Feedback and uses: Active filters, timers, multipliers, wave shaping

**EE**

Operational Amplifier, Application and Combinations of the Amplifier, Multivibrators

# Contents

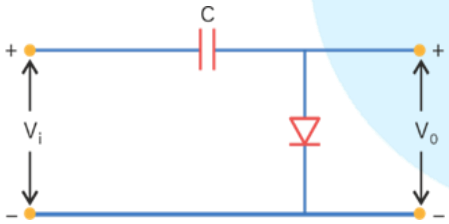

| Sr. | Chapters |
|---|---|
| 1. | Diode Family |
| 2. | Transistor Family |
| 3. | OP-AMP and its applications |
| 4. | BJT and MOSFET Amplifier |
| 5. | Feedback Amplifier and Oscillator |


■■■■■

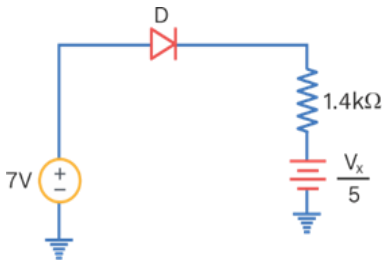
# 1 Diode Family

## MCQ, MSQ and NAT

1. For the circuit shown in the figure below, the input voltage is given as $V_i = 5 + 3\sin(\omega t)$ V. Considering the diodes to be ideal, the average output voltage is _____ Volts. (Rounded off to the nearest integer)




2. For the circuit shown in the figure below, the value of $V_x$ at which there is no flow of current in the given circuit will be _____ V. (Up to two decimal places). [Consider the potential barrier of the diode as 0.7 V]




3. A circuit consists of a DC power supply and has a full load voltage of 42 V and a no-load voltage of 50 V. The full load current is given as 2 A. The voltage regulation and output resistance respectively are –

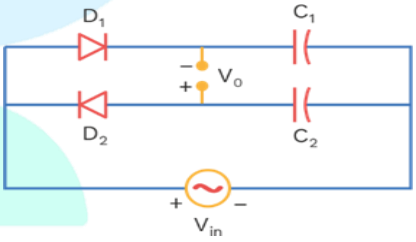
(A) 16 %, 4 Ω

(B) 19.04 %, 4 Ω

(C) 9 %, 8 Ω

(D) 16.2 %, 8 Ω

4. A circuit is shown in the figure below.




If the input to the circuit is alternating, then the given circuit is a -

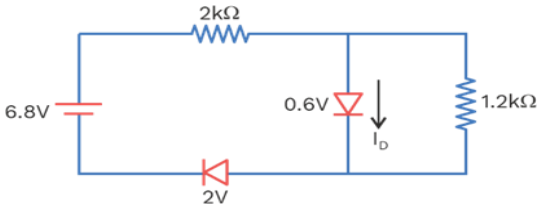
(A) Clamper circuit

(B) Voltage doubler circuit

(C) Clipper circuit

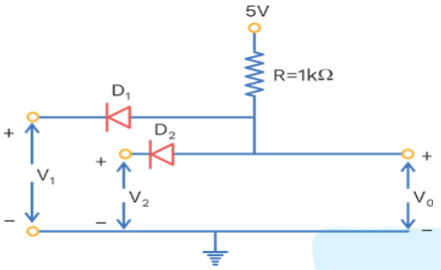
(D) Rectifier circuit

5. The circuit is shown in the figure below.

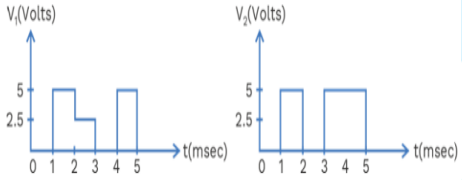



The current '$I_D$' through the diode is _____ mA. (Up to one decimal place)

6. The diodes shown in the circuit below are ideal.




The characteristics of the inputs $V_1$ and $V_2$ are shown below.




Which of the following statement(s) is/are true?

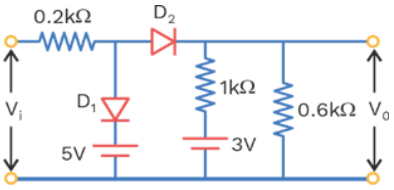
(A)The voltage $V_0$ = 5 V for time 0 < t < 1 msec.

(B)The current through the resistance is 2.5 mA for time 2 < t < 3 msec.

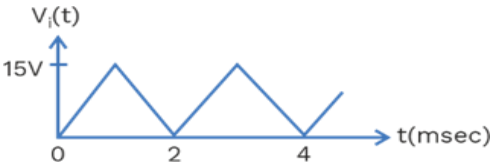
(C)The output voltage for 1 < t < 2 msec is $V_0$ = 5 V

(D)The current through the resistance is 5 mA for 3 < t < 4 msec.

7. A circuit is shown in the figure below.




The input voltage $V_i$(t) is-




Which of the following statement(s) is/are true?

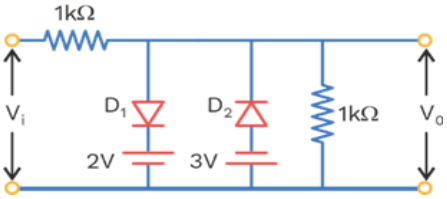
(A)The output voltage ($V_0$) when both the diodes are OFF is 1.125 V

(B)The maximum value of output voltage is $V_0$ = 5 V

(C)The input voltage at which the output voltage reaches its maximum value is 10.666 V

(D)The maximum value of output voltage is $V_0$ = 15 V

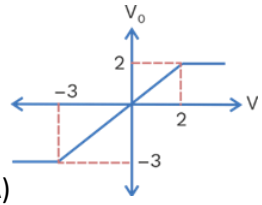
8. A circuit is shown in the figure below. The diodes $D_1$ and $D_2$ are ideal.



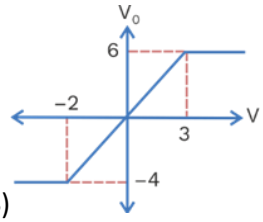

The transfer characteristics of the circuit is-

(Assume that $V_o$ and $V_i$ are expressed in volts)

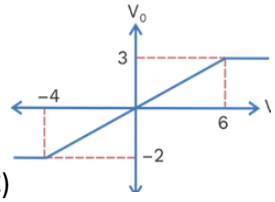
(A)



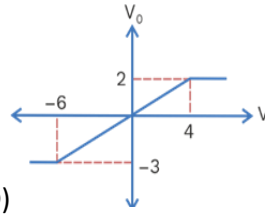

(B)



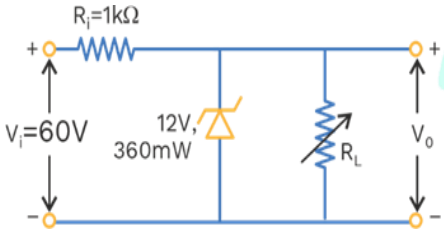

(C)




(D)




9. A Zener regulator is shown in the circuit below. Choose the correct options.



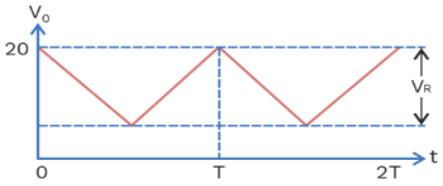

(A)$I_{Lmin}$ = 0 mA

(B)$I_{Lmax}$ = 48 mA

(C)$R_{Lmin}$ = 0.666 kΩ

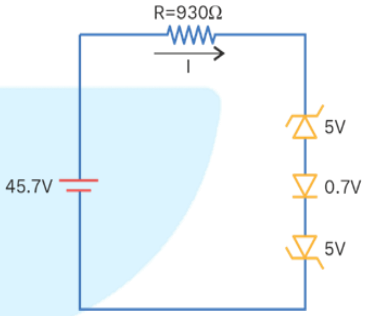
(D)$R_{Lmax}$ = 0.666 kΩ

10. A rectifier filter circuit has output voltage characteristics as shown below.



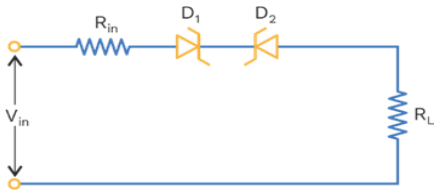

The peak-to-peak ripple is 10% of the peak output voltage. The average DC value of the output voltage($V_0$) is _____V. (Rounded off to the nearest integer)

11. The diodes shown in the circuit below are identical in construction.




The dynamic forward resistance = 5Ω and the dynamic reverse breakdown resistance = 10Ω. The current 'I' is _____ mA (Up up to 3 decimal places)

12. A circuit is shown in the figure below.




The input $V_{in}$ = 5 V square wave. The resistance $R_{in}$ = $R_L$ = 5 Ω. The Zener voltage $V_{z1}$ = 4V and $V_{z2}$ = 2V. The Zener diodes have negligible forward voltage drop and the diodes don't burn if 10mA <$I_z$< 400mA. The diode(s) for which regulation is preserved is/are?

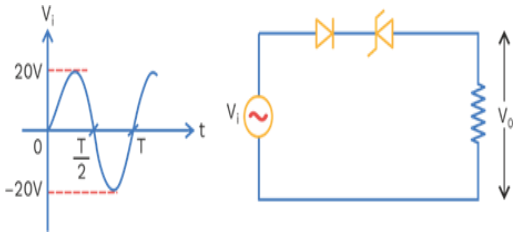
(A)Only $D_1$

(B)Only $D_2$

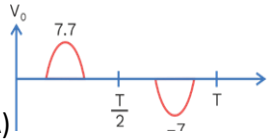
(C)Both $D_1$ and $D_2$

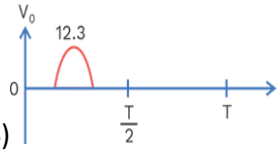
(D)Neither $D_1$ nor $D_2$

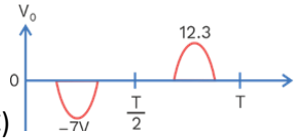
13. A circuit is shown in the figure below, where $V_z$ = 7V and the cut-in voltage of the diode is 0.7V.

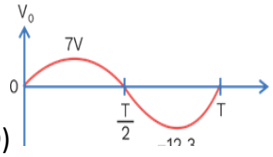



The output $V_0$ is –

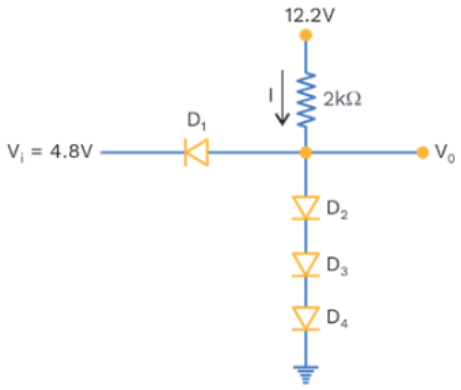
(A)



(B)



(C)



(D)



14. Identical diodes with the cut-in voltage of 0.8V are aligned as shown in the figure below.




$V_0$ and I are respectively –

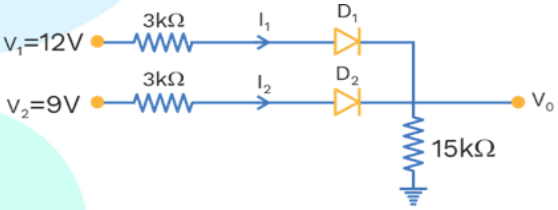
(A)0 V, 6.1 mA

(B)12.2 V, 0 mA

(C)2.1 V, 5.05 mA

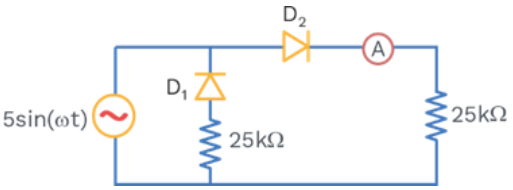
(D)2.4 V, 4.9 mA

15. For the circuit shown in the figure below, the cut-in voltage of the diode is 0.7 V.




The voltage $V_0$ is _____ Volts (Up to two decimal places)

16. The circuit shown below contains two ideal diodes and an ammeter that reads the average valve. The reading of the ammeter will be-



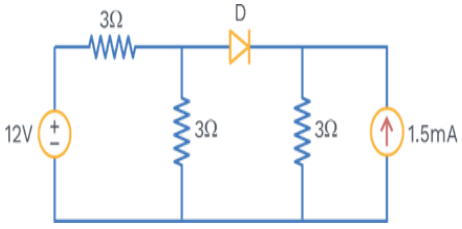

(A)$\frac{5}{\pi}$ mA

(B) $\frac{0.1}{\pi}$ mA

(C)$0.2\ \text{mA}$ (D) $\frac{0.2}{\pi}\ \text{mA}$

17. A circuit is shown in the figure below. The cut-in voltage of the diode is 0.7 V and the forward resistance is $0.5\ \Omega$. The current through the diode is _____ A.

(Up to 2 decimal places)



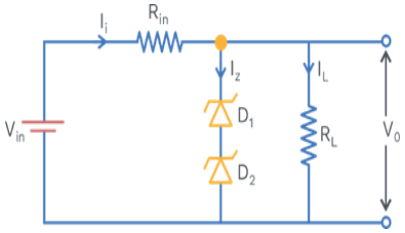

18. A list of specifications is given for the circuit shown below.

**DiodeD1 :**

$V_{z_1} = 7.2\ \text{V}\ for\ 25\ \text{mA} \le I_{z_1} \le 290\ \text{mA}$

**DiodeD2:**

$V_{z_2} = 7.7\ \text{V}\ for\ 2.3\ \text{mA} \le I_{z_2} \le 250\ \text{mA}$




If $5\ \text{mA} \le I_L \le 75\ \text{mA}$, then the range of $I_i$ for the safe operation of the diodes that are operating in the breakdown region is –

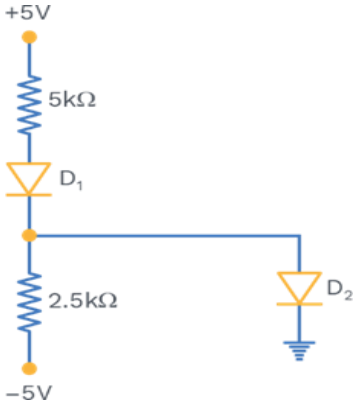
(A)$35\ \text{mA} \le I_i \le 250\ \text{mA}$

(B)$25\ \text{mA} \le I_i \le 250\ \text{mA}$

(C)$100\ \text{mA} \le I_i \le 255\ \text{mA}$

(D)$30\ \text{mA} \le I_i \le 250\ \text{mA}$

19. The diodes $D_1$ and $D_2$ shown in the figure below are ideal.




Which of the following statement(s) is/are true?

(A)The current through both the diodes $D_1$ and $D_2$ is 1 mA.

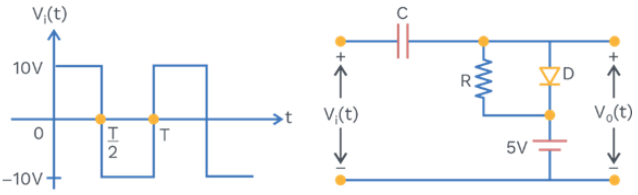
(B)The current through $D_1$ is 1 mA.

(C)The current through $D_1$ is 1.33 mA.

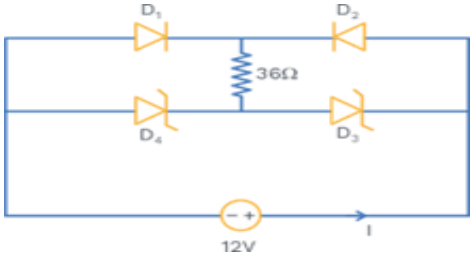
(D)The current through $D_2$ is 0 mA.

20. For the circuit shown in the figure below, the diode D is ideal and the resistance R is very high. The average value of the output voltage is_____ V.

(Rounded off to the nearest integer)



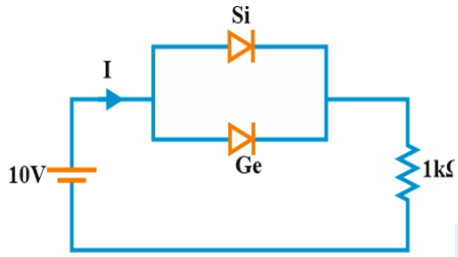

21. A circuit is shown below.



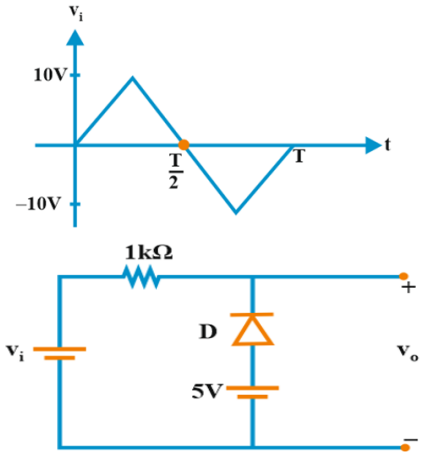
It is given that $V_Z$ = 7 V, Zener resistance = 10 Ω, $V_\gamma = 0.7\ V$ and forward resistance = 5 Ω.

The current I is _____ mA. (Up to 2 decimal places)

22. Find the current I (in mA) for the circuit as shown in the figure below. (Up to one decimal place)



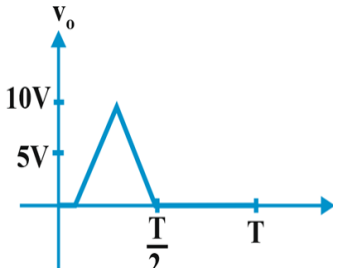

23. Consider the given diode circuit with input waveform as shown in the figure below.



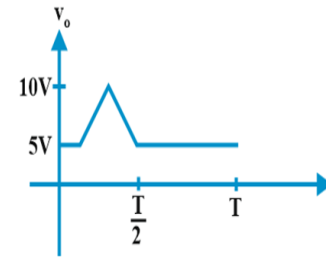

The output voltage waveform of the above circuit is-

(Consider that the diode D is ideal)

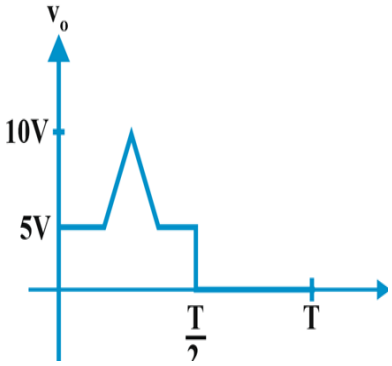
(A)



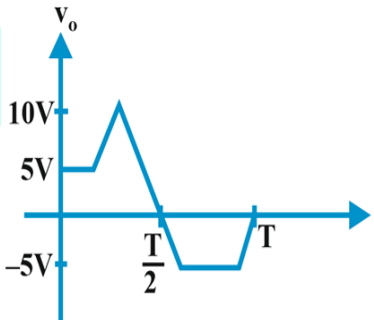

(B)



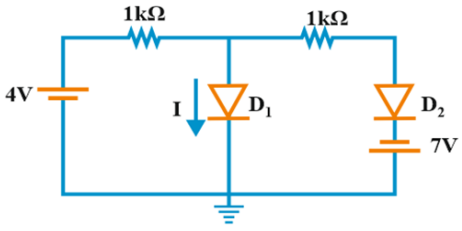

(C)




(D)




24. A diode circuit is given in the figure below. Consider all the diodes are ideal.

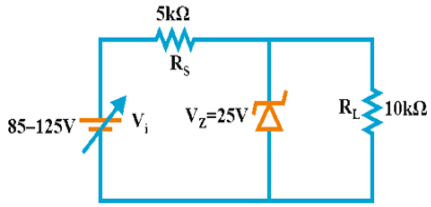



The current in the diode $D_1$ is-

(A)0 mA (B)1 mA

(C)3 mA (D)5 mA

25. For the Zener regulator circuit shown in the figure below, the maximum and minimum currents flowing through the Zener diode are respectively-




(A)17.5 mA and 7 mA

(B)20 mA and 17.5 mA

(C)17.5 mA and 9.5 mA

(D)20 mA and 12 mA

26. A full wave bridge rectifier has sinusoidal input of 120 V RMS value & a load resistance of 1 kΩ. Assuming that all the diodes are made of silicon with a cut in voltage of 0.7 V.

Which of the following statement(s) is/are correct?

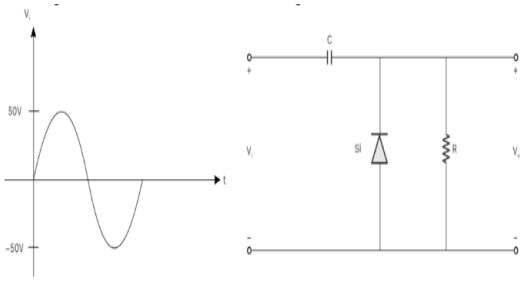
(A)PIV rating of each diode = 169 V

(B)Peak current, $I_{max}$ = 168.3 mA

(C)Peak current, $I_{max}$ = 149 mA

(D)Average voltage at load, $V_{DC}$ = 107.14 V

27. For the given circuit, assume diode cut-in voltage = 0.7 V.




Which of the following statement(s) is/are correct?

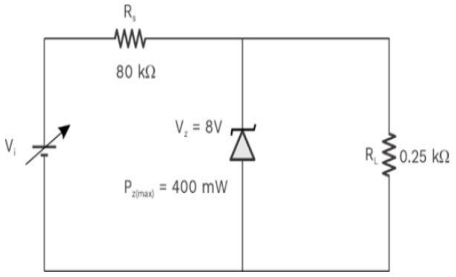
(A)For positive half cycle of input, output $V_0$ = 99.3 V

(B)For positive half cycle of input, output $V_0$ = -99.3 V

(C)For negative half cycle of input, output $V_0$ = 0.7 V

(D)For negative half cycle of input, output $V_0$ = -0.7 V

28. A Consider the circuit shown in the figure below.




Find the range of input voltage $V_i$ that will maintain the output voltage at 8V and does not exceed the maximum power rating of the Zener diode.

(A)$V_i \Rightarrow 10.56\ \text{V}$ to $14.56\ \text{V}$

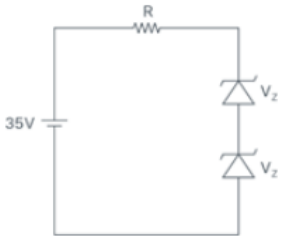
(B)$V_i \Rightarrow 8\ \text{V} to 12\ \text{V}$

(C)$V_i \Rightarrow 9.56\ \text{V}$ to $13.56\ \text{V}$

(D)None of these

29 A circuit is shown below, the voltage

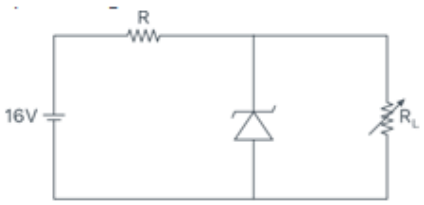
across each Zener diode is 10 Volt and current through Zenner diode is 150mA, the resistance (R) is-




(A)100 Ω (B)125 Ω

(C)25 Ω (D)160 Ω

30. Zener diode of rating 12V and 300mW is used for providing voltage stabilization to the supply of variable load. The minimum current of 5mA should flow through the Zener for proper Zener action. The resistance R and maximum load current if input voltage is 16V, are-



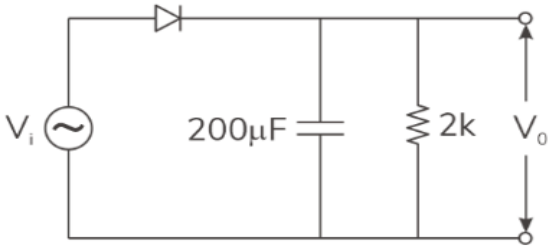

(A)$320\Omega, 15\ \text{mA}$

(B)$160\Omega, 15mA$

(C)$320\Omega, 20mA$

(D)$160\Omega, 20mA$

31. Consider the circuit shown below (diode is ideal) if the input voltage is $V_i = 50\sin(100\pi t)$. Find average dc output voltage and average diode current.



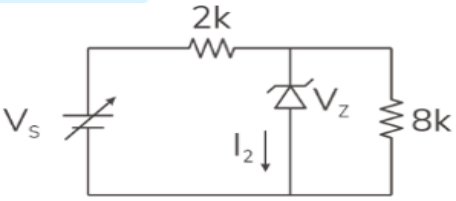

(A)$V_{dc} = 58.8\ \text{V}, \quad I_{dc} = 0.515\ \text{A}$

(B)$V_{dc} = 48.8\ \text{V}, \quad I_{dc} = 5.15\ \text{A}$

(C)$V_{dc} = 48.8\ \text{V}, \quad I_{dc} = 0.515\ \text{A}$

(D)$V_{dc} = 58.8\ \text{V}, \quad I_{dc} = 5.15\ \text{A}$

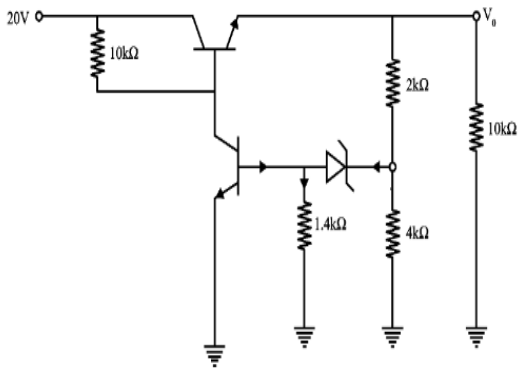
32. A Zener regulator circuit is shown below. It is known that Zener undergoes breakdown for supply voltage of $V_S = 10\ \text{V}$ and maximum power dissipation occurs at $V_S = 20V$. Find $I_{z\max} - I_{z\min}$ in mA.



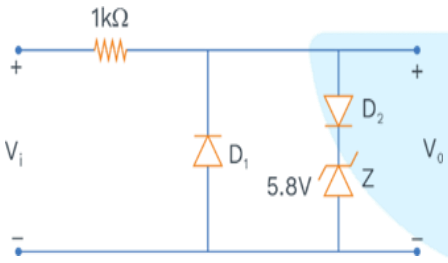

(A)10 (B)5

(C)9 (D)1

33. In the circuit shown above Zener break down voltage is 10 V. For both the transistors $V_{BE(on)} = 0.7\ \text{V}$ and $\beta$ is very high. Then the value of $V_0$ is _____Volts. Assume knee current of Zener diode to be 0.5mA.




34. The input voltage to the limiter circuit is given as $V_i = 5\sin(50\pi t)$ volts. The voltage drop across the diode is 0.7V in forward-biased condition.

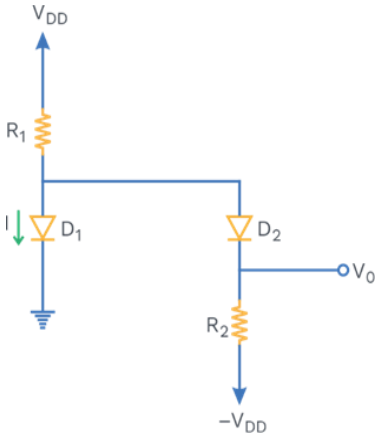



The maximum value of the output voltage is _____V.

(Up to one decimal place)

35. For the circuit shown below, find the value of current I.

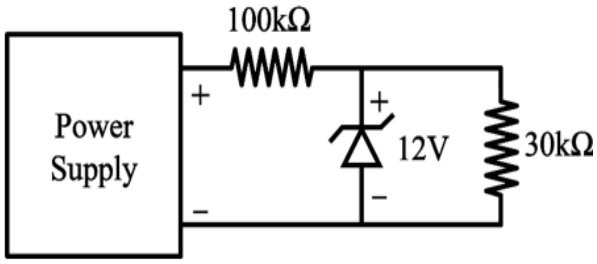
Given: $V_{DD} = 9\ \text{V}, \text{R}_1 = 2\text{k}\Omega, \text{R}_2 = 5\text{k}\Omega$



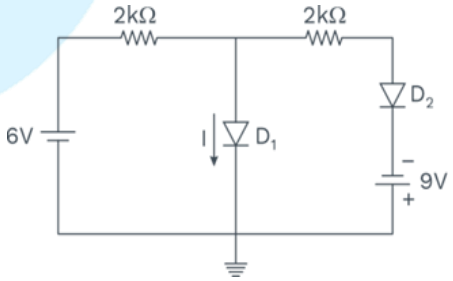

(A)1.2 mA (B)2.7 mA

(C)4.5 mA (D)3 mA

36. Assume that the supply voltage in the figure shown below starts decreasing from 60 V to 0 V. The maximum supply voltage at which the Zener diode loses its regulating ability is _____ V.



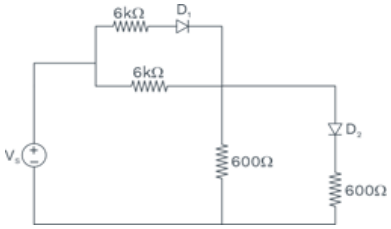

37. Assuming that the diodes are ideal as shown in figure, then the value of current in the diode $D_1$ is-




(A)– 3 mA (B)8 mA

(C)0 mA (D)6 mA

38. For the circuit shown below each diode has $\text{V}_\gamma = 0.6\ \text{V}$, and $r_\text{d} = 0\Omega$. Both diodes will be ON if

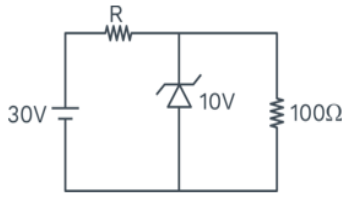



(A)$V_S$< 3.3 V (B)2 < $V_S$< 3

(C)$V_S$> – 3.6 V (D)$V_S$> 3.9 V

39. Figure shows an electronic voltage

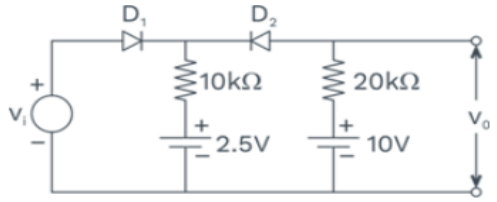
regulator. The zener diode may be assumed to require a minimum current of 25 mA for satisfactory operations. The value of R required for satisfactory voltage regulation of the circuit is




(A)$80\Omega$ (B)$160\Omega$

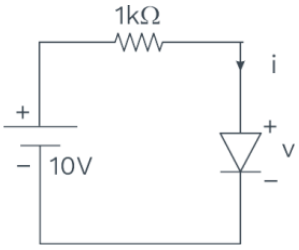
(C)$90\Omega$ (D)$40\Omega$

40. For the clipper circuit shown in the figure, the first break point occurs when $v_i$ is at _____ Volts.




41. The i–v characteristics of the diode in the circuit given below are

$$i=\begin{cases}\dfrac{v-0.7}{500}A, & v\geq 0.7V\\ 0\ A, & v<0.7V\end{cases}$$

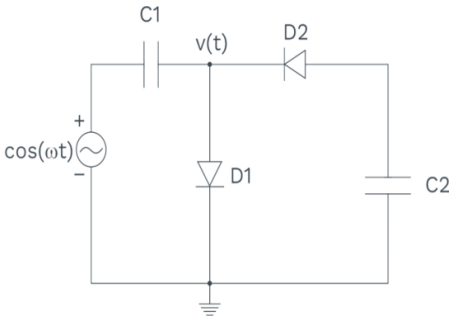



The current in the circuit is

(A) 10 Ma (B) 9.3 mA

(C) 6.67 mA (D) 6.2 mA

42. The diodes and capacitors in the circuit shown are ideal. The voltage $v(t)$ across the diode D1 is



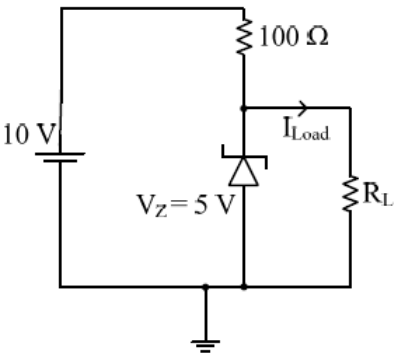

(A) $\cos(\omega t)-1$

(B) $\sin(\omega t)$

(C) $1-\cos(\omega t)$

(D) $1-\sin(\omega t)$

43. In the circuit shown below, the knee current of the ideal Zener diode is 10mA. To maintain 5 V across $R_L$, the minimum value pf $R_L$ in $\Omega$ and the minimum power rating of the Zener diode in mW, respectively, are




(A) 125 and 125

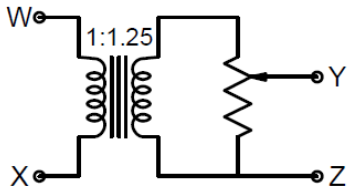
(B) 125 and 250

(C) 250 and 125

(D) 250 and 250

44. The following arrangement consists of an ideal transformer and an attenuator

which attenuates by a factor of 0.8. An ac voltage $V_{WX1} = 100\,V$ is applied across WX to get an open circuit voltage $V_{YZ1}$ across YZ. Next, an ac voltage $V_{YZ2} = 100\,V$ is applied across YZ to get an open circuit voltage $V_{WX2}$ across WX. Then, $V_{YZ1} / V_{WX1} / V_{WX2} / V_{YZ2}$ are respectively,



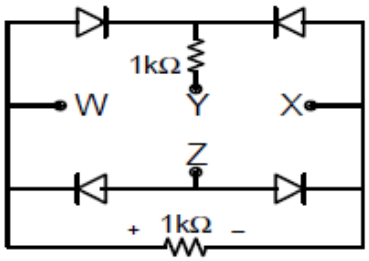

(A) 125/100 and 80/100

(B) 100/100 and 80/100

(C) 100/100 and 100/100

(D) 80/100 and 80/100

45. A voltage $1000 \sin \omega t$ Volts is applied across YZ. Assuming ideal diodes, the voltage measured across WX in Volts, is




(A) $\sin \omega t$

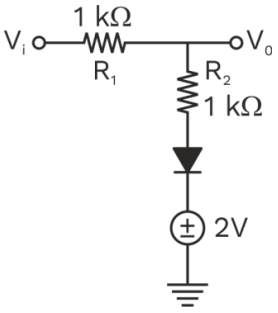
(B) $\left(\sin \omega t + |\sin \omega t|\right) / 2$

(C) $\left(\sin \omega t - |\sin \omega t|\right) / 2$

(D) 0 for all $t$

46. The diode in the circuit shown has $V_{on} = 0.7$ Volts but is ideal otherwise.

If $V_i = 5 \sin(\omega t)$ Volts, the minimum and maximum values of $V_o$ (in Volts) are, respectively,



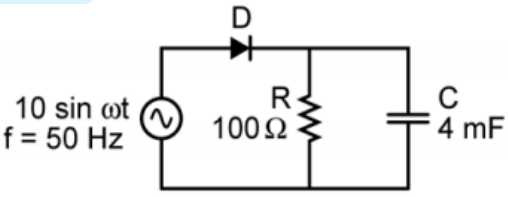

(A) –5 and 2.7

(B) 2.7 and 5

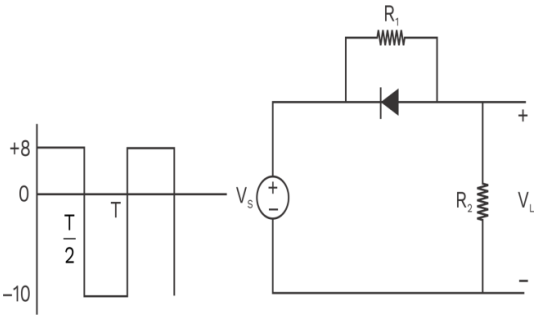
(C) –5 and 3.85

(D) 1.3 and 5

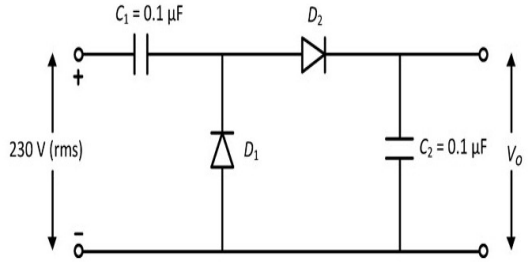
47. The figure shows a half-wave rectifier. The diode D is ideal. The average steady state current (in Amperes) through the diode is approximately…….




48. In the circuit shown, $V_S$ is a square wave of period T with maximum and minimum values of 8V and –10V, respectively. Assume that the diode is ideal and $R_1 = R_2 = 50\Omega$. The average value of $V_L$ is _____ volts (rounded off to 1 decimal place).




49. In the circuit shown below, all the components are ideal and the input voltage is sinusoidal. The magnitude of the steady-state output $V_0$ (rounded off to two decimal places) is _____ V.



# Answer Key

Q.1 Ans. [–3]

Q.2 Ans. [31.5]

Q.3 Ans. [B]

Q.4 Ans. [B]

Q.5 Ans. [ 1.6]

Q.6 Ans. [C, D]

Q.7 Ans. [ A ,B ,C]

Q.8 Ans. [ D ]

Q.9 Ans. [ B ,D ]

Q.10 Ans. [19]

Q.11 Ans. [ 41.368 ]

Q.12 Ans. [ C ]

Q.13 Ans. [ B ]

Q.14 Ans. [ D ]

Q.15 Ans. [ 9.41]

Q.16 Ans. [ D ]

Q.17 Ans. [ 0.16]

Q.18 Ans. [ C ]

Q.19 Ans. [ C ,D ]

Q.20 Ans. [ -5 ]

Q.21 Ans. [ 84.31]

Q.22 Ans. [ 9.7]

Q.23 Ans. [B]

Q.24 Ans. [ A ]

Q.25 Ans. [ C ]

Q.26 Ans. [ A ,B ,D ]

Q.27 Ans. [ A ,D ]

Q.28 Ans. [ A ]

Q.29 Ans. [ A ]

Q.30 Ans. [ D ]

Q.31 Ans. [ C ]

Q.32 Ans. [ B ]

Q.33 Ans. [ 17.05 ]

Q.34 Ans. [ 6.5 ]

Q.35 Ans. [ B ]

Q.36 Ans. [ 52 ]

Q.37 Ans. [ C]

Q.38 Ans. [ D ]

Q.39 Ans. [ B ]

Q.40 Ans. [ 5 ]

Q.41 Ans. [D]

Q.42 Ans. (A)

Q.43 Ans. (B)

Q.44 Ans. [C]

Q.45 Ans. [D]

Q.46 Ans. [C]

Q.47 Ans. [0.09, Range 0.08 to 0.12]

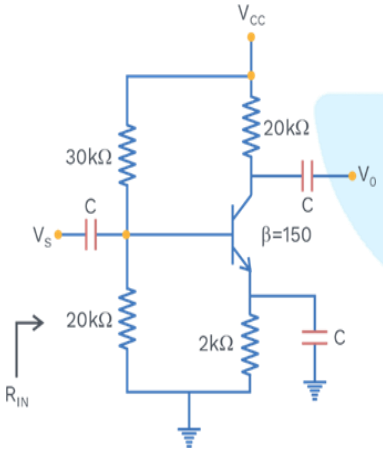
Q.48 Ans. [-3, Range -3.1 to -2.9]

Q.49 Ans.[650.53, Range 644 to 657]

# 2 Transistor Family

## MCQ, MSQ and NAT

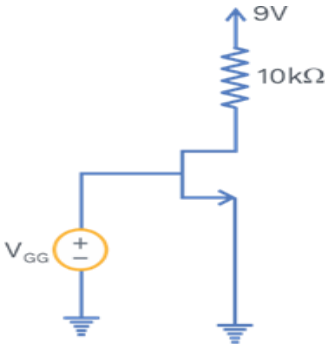
1. For the transistor circuit given below, if the transconductance of the transistor is 25 mS, then the value of input impedance is_____ kΩ. (Rounded off to the nearest integer)




2. For the circuit shown in the figure, the transistor has threshold voltage ($V_T$) = 5 V, the transconductance parameter

$$(K_n) = \mu_0 C_{ox}\left(\frac{W}{L}\right) = 2\ \text{mA/V}^2$$

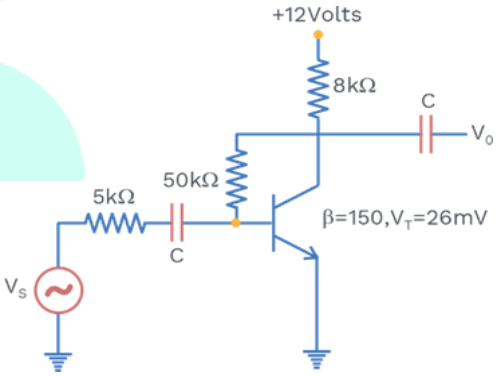
and the channel length modulation parameter $(\lambda) = 0$. The transistor is operating in the saturation region. If $I_D$ is to be 0.6 mA, then the value of $V_{GS}$ is-




(A)2.9 V (B)3.3 V

(C)5.8 V (D)3.6 V

3. The voltage gain $\left(\left|\frac{V_0}{V_s}\right|\right)$ of the circuit shown below is-



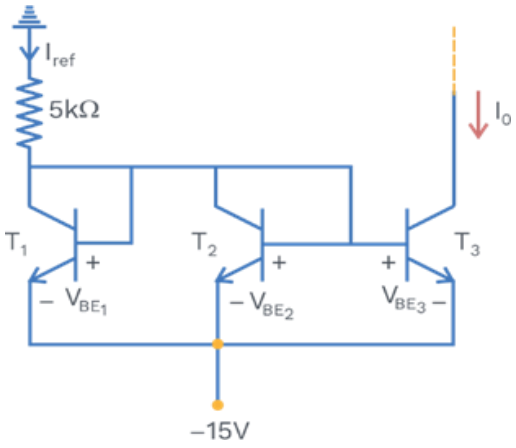

[Consider that $V_{BE}$=0.7 V]

(A)15.62 (B)9.25

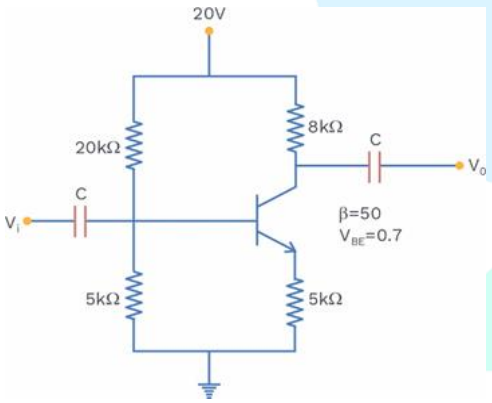
(C)100 (D)36.33

4. A current mirror circuit is given below. The base to emitter voltage of the transistors $V_{BE_1} = V_{BE_2} = V_{BE_3}$ = 0.7 V and β= 30. If all the transistors are identical, then the current $I_0$ is_____

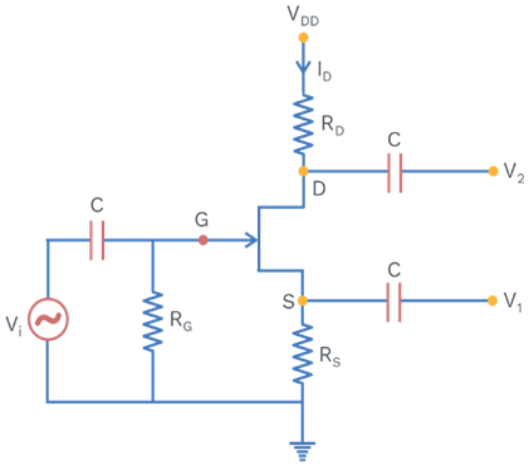
mA. (Up to three decimal places)




5. A BJT configuration is given below. The value of $h_{ie}$ is_____ Ω (Assume $V_T$ = 26 mV) (Up to two decimal places)



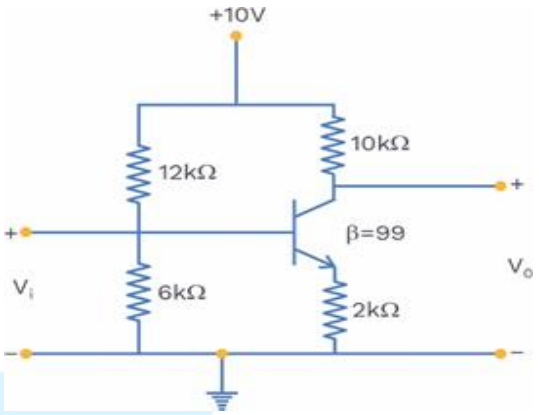

6. In the given FET circuit, which one of the given relations is correct?




(A) $V_2 = \frac{-R_D}{R_S}V_1$ (B) $V_2 = \frac{R_S}{R_D}V_1$

(C) $V_2 = \frac{R_D}{R_s}V_1$ (D) $V_2 = \frac{-R_S}{R_D}V_1$

7. Which of the parameter(s) is/are correct for the given CE amplifier circuit?



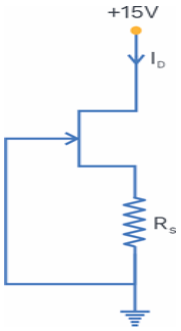

(A)Stability factor, S = 100

(B)$V_{Th}$ = 3.3 V

(C)Stability factor, S = 25.6

(D)Stability factor, S = 2.94

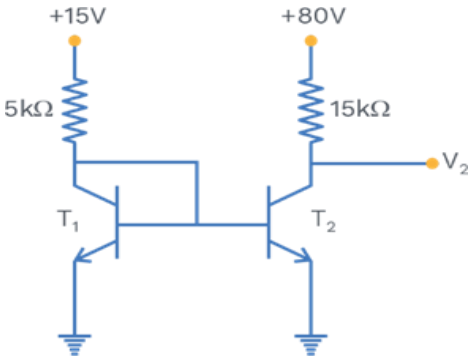
8. The given FET circuit has $I_{DSS}$ = 15 mA and $V_P$ = -2V. Find the value of $R_s$ in the given circuit, if $I_D$ = 5.5 mA.




(A)$126.05\Omega$ (B)$143.6\Omega$

(C)$312.5\Omega$ (D)$63.77\Omega$

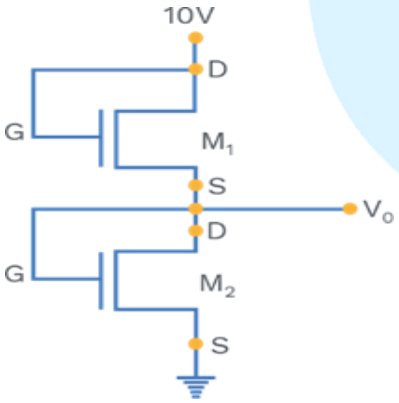
9. The matched transistors $T_1$ and $T_2$ are shown in the figure. If β = 50 and $V_{BE}$ = 0.7, then-




(A)$I_{C_2} = 2.75\ mA$ (B)$V_0$ = 38.75 Volts

(C)$V_0$ = 21.62 Volts (D)$I_{C_2} = 2.86\ mA$

10. The MOSFETs shown in the circuit are identical. If $V_T$ = 0.8 V, $\mu_n C_{ox}$ = 30 μA/Volt² and $\left(\frac{W}{L}\right) = 60$, then-



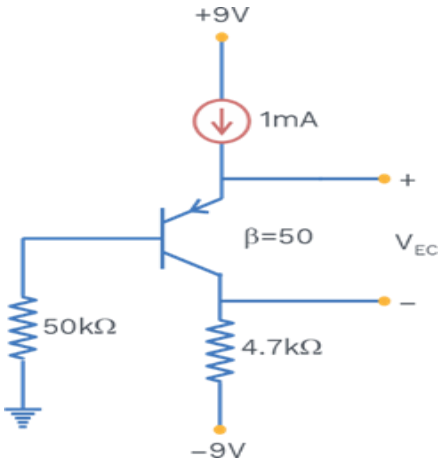

(A)$V_0$ = 4.2 V

(B)$V_0$ = 5V

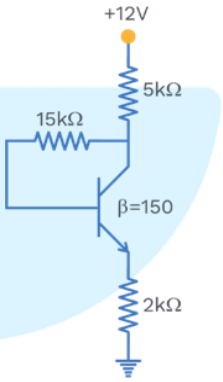
(C)$I_D$ = 17.655 mA

(D)$I_D$ = 15.876 mA

11. A PNP transistor circuit is given below. Find the voltage $V_{EC}$. (Assume $V_{EB}$ = 0.7 V and β = 50) (Up to two decimal places)



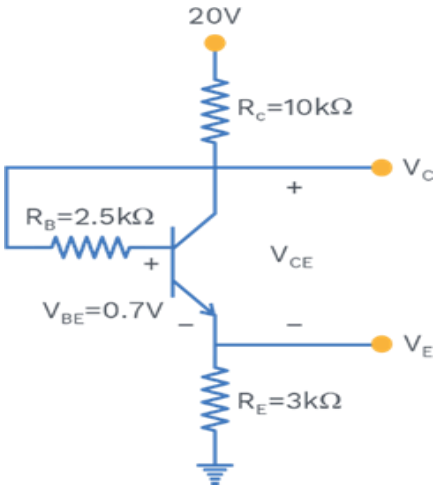

12. Find the stability factor of the given collector to base bias circuit.




(A)5.5 (B)3.1

(C)39.2 (D)151

13. The given transistor has β = 49. Which of the parameter(s) is/are correct for the transistor?



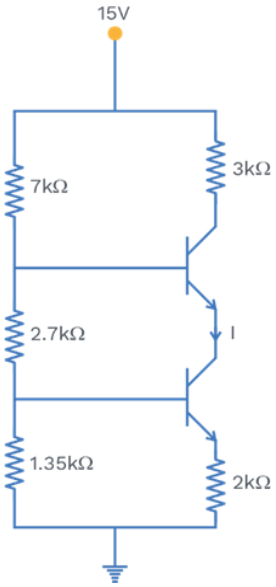
(A)Q-point of the transistor is (0.765 V, 1.45 mA)

(B)Q-point of the transistor is (0.525 V, 2.95 mA)

(C)$V_B$ = 5.14 Volts

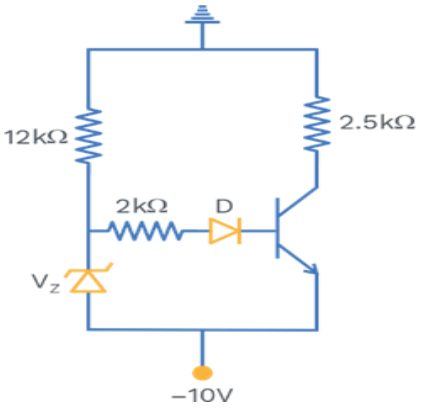
(D)$V_C$ = 5.2 Volts

14. An In the circuit shown below, the two BJTs are biased in the active region and have a very high value of β. If $V_{BE}$ = 0.6 V (for both BJTs), then the value of current I is_____ mA. (Up to three decimal places)




15. The Voltage regulator transistor circuit is shown below. The Zener diode voltage $V_Z$ = 8 Volts, voltage across diode D = 0.7 V and β = 60. If the Zener diode is in the breakdown, then the collector current is _____mA. (Up to two decimal places)

[Assume that $V_{BE}$ = 0.7 V in active mode and $V_{BE}$ = 0.8 V in saturation mode]




16. An n–channel MOSFET has the following parameters.

$$k_n = \mu_n C_{ox}\left(\frac{W}{L}\right) = 20\mu A/V^2$$

Threshold Voltage ($V_{TN}$) = 1 V

Channel length modulation parameter (λ) = 0

Gate to drain capacitance ($C_{gd}$) = 0.05 pF

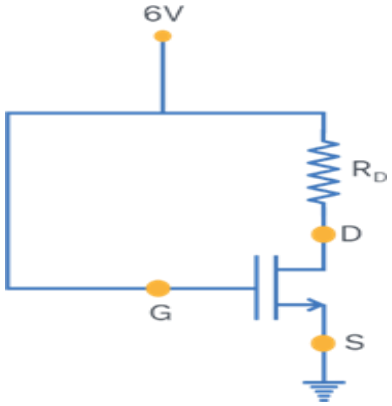
Gate to source capacitance ($C_{gs}$) = 0.3 pF

If the transistor is biased at $V_{GS}$ = 6V, then the unity-gain bandwidth of the transistor will be -

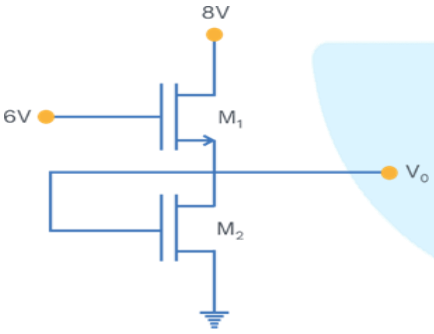
(A)323.67 kHz (B)12.52 MHz

(C)127.5 kHz (D)90.94 MHz

17. The MOS transistor has threshold voltage $V_T$ = 2V, drain voltage $V_{DS}$ = 1V, $\mu_n C_{ox}$ = 3 μA/$V^2$ and $\left(\frac{W}{L}\right) = 3$. The drain resistance ($R_D$) for the transistor is _____ kΩ. (Up to two decimal places)




18. Two MOS transistors $M_1$ and $M_2$ given below have $k_n' = \mu_n C_{ox} = 0.2$ mA/V$^2$, $V_T = 1$ V. If $\left(\frac{W}{L}\right)_1 = 5$ and $\left(\frac{W}{L}\right)_2 = 2$, then-




(A)$V_{GS}$ of $M_1$ = 2.55 Volts

(B)$V_0$ = 3.45 Volts

(C)$V_{GS}$ of $M_1$ = 4.08 Volts

(D)$V_0$ = 1.92 Volts

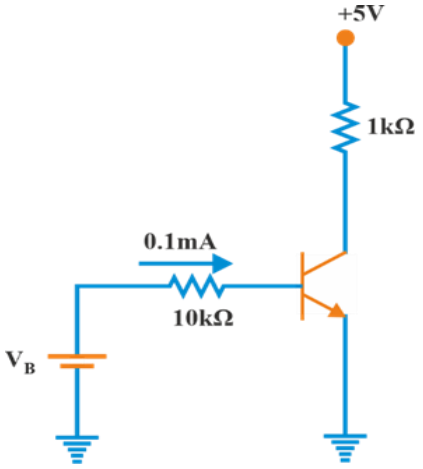
19. A transistor has a current gain of 0.95 in the common base configuration. Find the current gain in the common collector configuration.

(A)50 (B)20

(C)40 (D)100

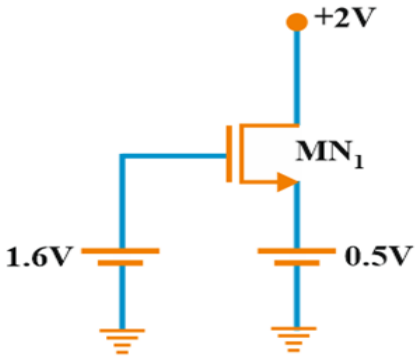
20. Consider the transistor circuit shown in the figure below.

Assume $V_{CE(sat)} = 0.2$ V




The minimum value of β, such that the transistor is in the saturation region, is _____. (Rounded off to the nearest integer)

21. A MOS circuit is shown in the figure below. Consider the threshold voltage, $V_T = 0.6$ V




The region of operation of the transistor $MN_1$ is –

(A)Cut-off region

(B)Saturation region

(C)Linear region

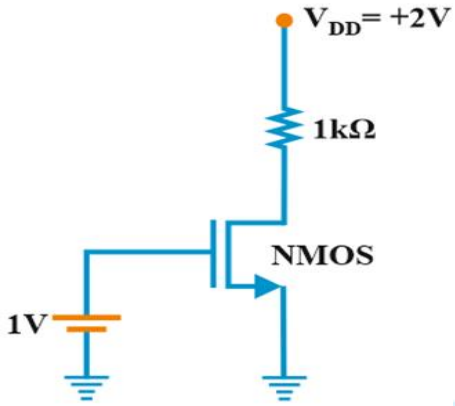
(D)Cannot be determined

22. A MOSFET circuit is shown in the figure below. Consider the transistor parameters as follows.

Threshold Voltage, $V_T = 0.5\ V$

$\mu_n C_{ox} = 300\mu A/V^2$

If the transistor operates at the edge of saturation, then the $\frac{W}{L}$ ratio is –

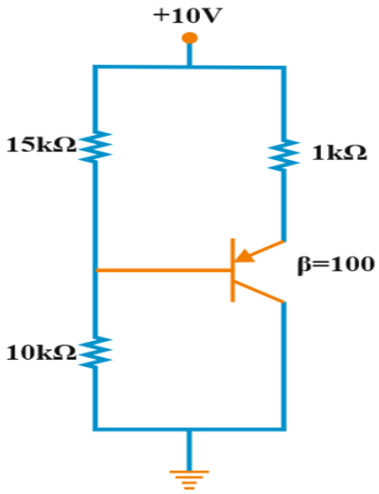



(A)10 (B)20

(C)30 (D)40

23. For the given transistor circuit, the base voltage $V_B$ is –

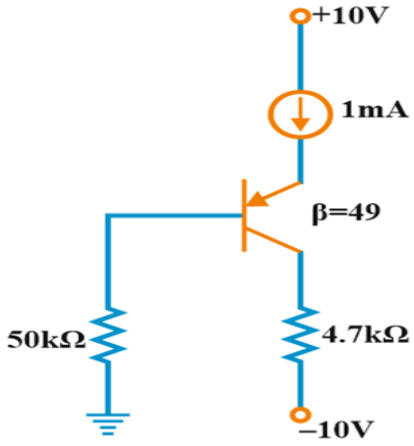
(Take $V_{EB}$ = 0.7 V)




(A)3.162 V (B)3.927 V

(C)4.294 V (D)5.692 V

24. Consider the transistor circuit shown below.

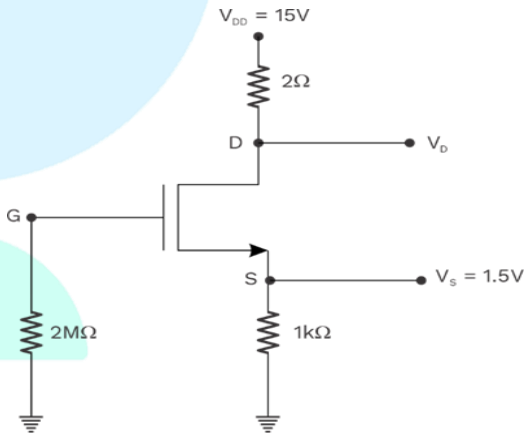



The value of voltage $V_{EC}$ is –

(A)7.094 V (B)3.135 V

(C)5.182 V (D)4.243 V

25. A MOSFET biasing circuit is given below.




Consider pinch off voltage, $V_P$ = -4 V.

Which of the following statement(s) is/are correct?

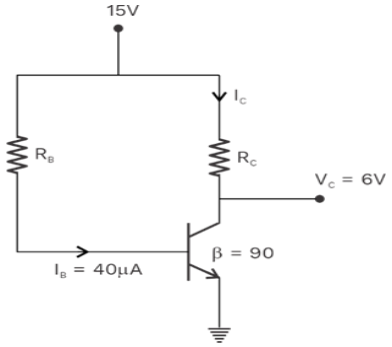
(A)Maximum drain current, $I_{DSS}$ = 3.85 mA

(B)Drain voltage, $V_D$ = 12 V

(C)Drain to source voltage, $V_{DS}$ = 13.5 V

(D)Drain to source voltage, $V_{DS}$ = 10.5 V

26. Consider the fixed bias circuit shown in the figure below.

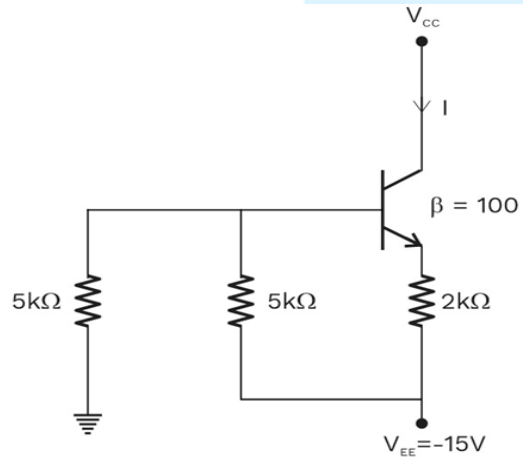



Assuming $V_{BE}$ = 0.7 V, find the value of resistance $R_C$.

(A)$2.5\Omega$ (B)$2.5\text{k}\Omega$

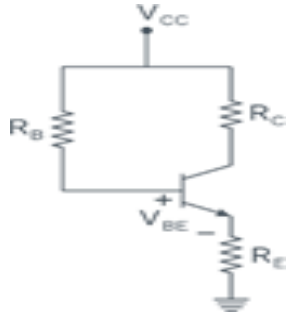
(C)$250\text{k}\Omega$ (D)None of these

27. For the circuit shown below consider $V_{BE} = 0.7\ \text{V}$




The value of current I is _____ mA.

28. In the circuit shown below, find the stability factor when $R_B = R_C = R_E = R$.




(A)$S = (1+\beta)$

(B)$S = \dfrac{2(\beta+1)}{(\beta+2)}$

(C) S = 1

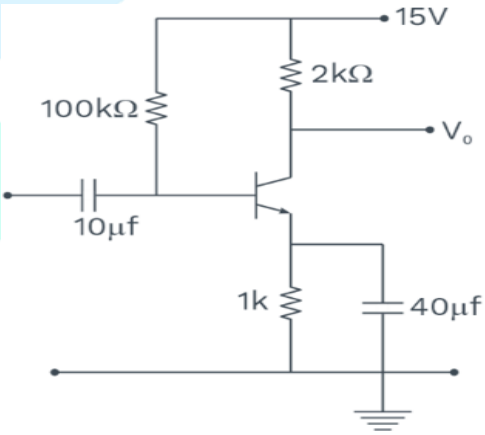
(D)$S = \dfrac{3(\beta+1)}{(\beta+3)}$

29. For a transistor amplifier with self biasing network, the following components are used $R_1$ = 4kΩ, $R_2$ = 4kΩ ,$R_E$ = 1kΩ and $\beta$ = 60 . The approximate value of the stability factor 'S' will be:

(A)4 (B)3

(C)2 (D)1.5

30. The circuit using a BJT using $\beta = 40$ and $\text{V}_{\text{BE}} = 0.7\ \text{V}$ is shown in the figure. The base current $I_B$ and output voltage $V_0$ are respectively.



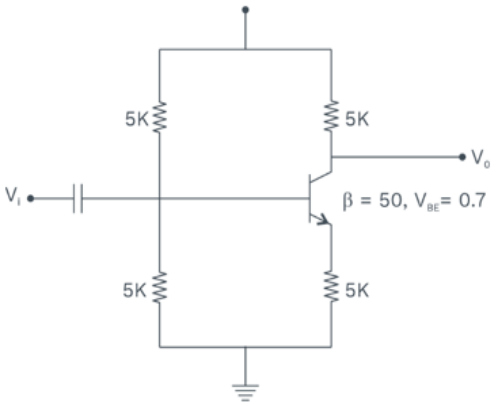

(A)$I_B = 1\ \text{mA},\ \text{V}_0 = 7\ \text{V}$

(B)$\text{I}_\text{B} = 0.1\ \text{mA},\ \text{V}_0 = 7\ \text{V}$

(C)$I_B = 2.5\ \text{mA},\ \text{V}_0 = 5\ \text{V}$

(D)$I_B = 0.25\ \text{mA},\ \text{V}_0 = 5\ \text{V}$

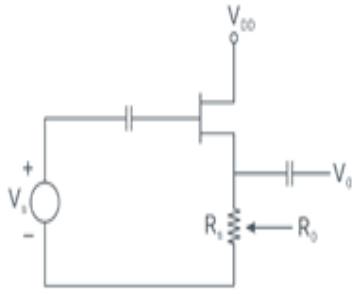
31. Consider a transistor connected as a common-emitter amplifier as shown in the figure. The quiescent collector voltage of the circuit is approximately –




(A)8.22 volts (B)12 volts

(C)7.3 volts (D)5.5 volts

32. The circuit is shown in the figure below

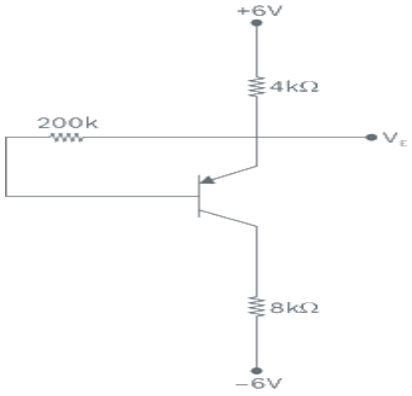



If $g_m = 2 \times 10^{-3} S$ and $R_s = 2000\Omega$; then $R_0$ is -

(A)100 $\Omega$ (B) $200\Omega$

(C)$300\Omega$ (D)$400\Omega$

33. A circuit is shown in the figure below

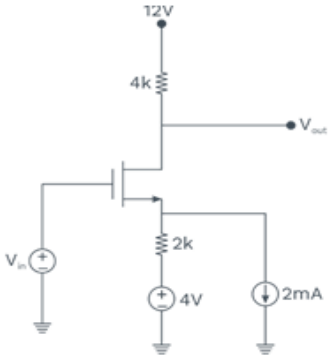



If the voltage $V_E$ is given as 4V, then the value of $\beta$ is-

(A)20 (B)17.86

(C)18.8 (D)15

34. A circuit is shown is the figure below.

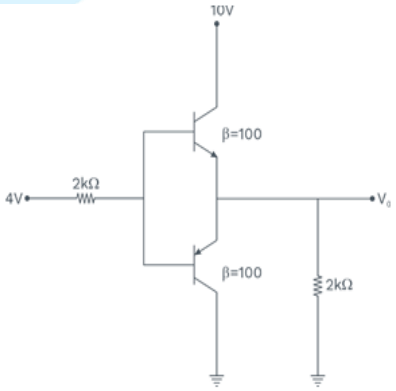



The voltage $V_T$ is given as 1V and K is very high. If the input voltage is 7V, then the output voltage is -

(A)6V (B)4V

(C)3V (D)1V

35. A circuit is shown in the figure below.



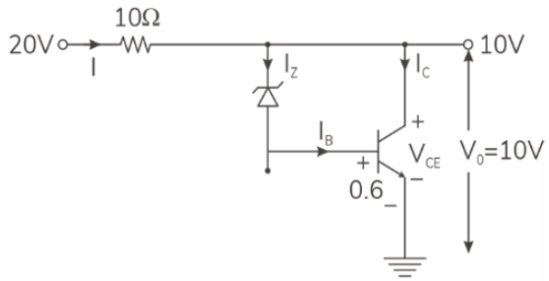

The output voltage $V_0$ of the given circuit is

(A)5V (B)5.2 V

(C)4.5V (D)3.3V

36. Consider a shunt regulator circuit as shown below has a variable unregulated supply of (10 – 20)V. BJT has $\beta = 99$ and $V_{BE} = 0.6$ V. Neglect the current

through $R_B$, if maximum power dissipation of Zener is $P_2$ and that of transistor is $P_T$, then for $P_T = xP_Z$. What is the value of x?

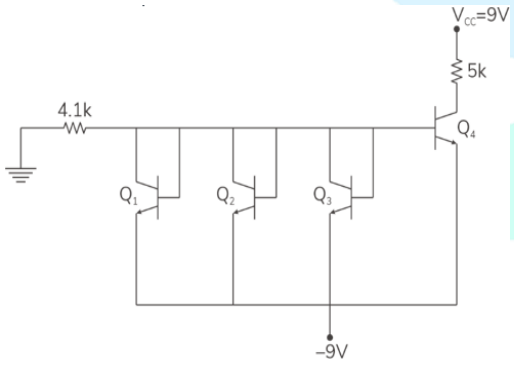



(A)100 (B) 133.3

(C)105.3 (D)110.5

37. Consider the current mirror circuit. All transistors have $(V_{BE})_{in} = 0.8$ and $\beta = 100$.

Then choose the correct option.




(1) $I_{E_4} = 0.66\ \text{mA}$

(2) $I_{C_2} = 0.658\ \text{mA}$

(3) $I_{B_1} = 3.2\mu\text{A}$

(4) $I_{B_3} = 6.58\mu\text{A}$

Which of the following parameters are correct?

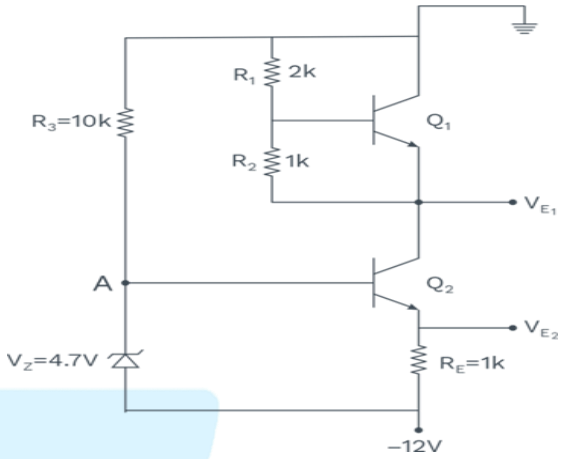
(A)1, 2 and 3 only

(B)2, 3 and 4 only

(C)1, 2 and 4 only

(D)1, 3 and 4 only

38. Consider the BJT circuit shown below with transistor having $V_{BE}$ = 0.7V and large β. Determine the region of operation of transistor '$Q_1$'.



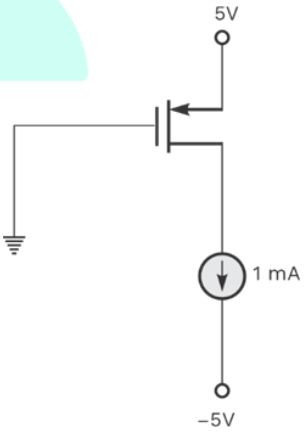

(A)Active

(B)Cut off

(C)Saturation

(D)Insufficient data

39. Consider the circuit shown below.




The parameters for N-MOS are given as

$\frac{\mu_n C_{ox} W}{2L} = \frac{1}{2}\ \text{mA/V}^2$

$V_T$ = 1.2 V

λ = 0

Which of the following statement(s) is/are true?

(A)$V_{DS} > V_{GS} - V_T$

(B)$V_{DS} < V_{GS} - V_T$

(C)$V_{DS} = 7.614V$

(D)$V_{DS} = 2.614V$

40. The figure shown below represents a MOS transistor for small analysis,

where $K_n = \frac{\mu_n C_{ox} W}{2L} = 10\ \text{mA/V}^2$.

MOSFET is biased at a DC current, $I_D$ = 10mA and small signal input voltage of $V_{sig}$ = 1mV.

Which of the following statement(s) is/are true?

(Consider $\lambda = 0$)

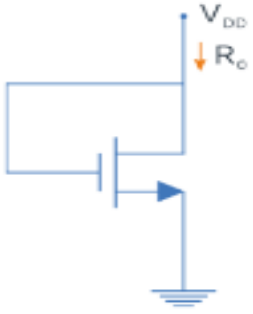
(A)$g_m = 10$ mA/V

(B)$g_m = 20$ mA/V

(C)$V_0 = -0.2V$

(D)$V_0 = -0.4V$

41. Consider the given MOSFET circuit shown below. Calculate the resistance $R_0$ seen at the drain terminal, if $\lambda \neq 0$



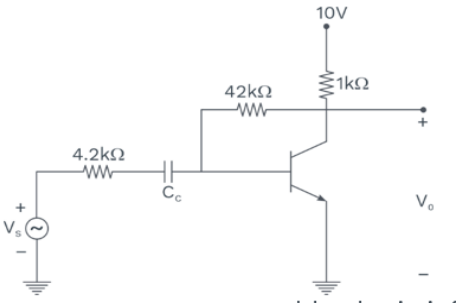

(A)$r_0$

(B)$\frac{1}{g_m} + r_0$

(C)$\frac{g_m r_0}{1 + g_m}$

(D)$\frac{r_0}{1 + g_m r_0}$

42. A circuit is shown in the figure below




The transistor has $\beta_{DC} = 50$, $V_{BE}$ = 0.6V and $h_{ie}$, $h_{fe}$ is infinite. If the capacitance in the circuits is also finite, then which of the following statement(s) is/are TRUE?

(A)The voltage $V_{CE}$ is when considering DC condition is 4.85V

(B)The voltage $V_{CE}$ when considering DC conditions is 4V

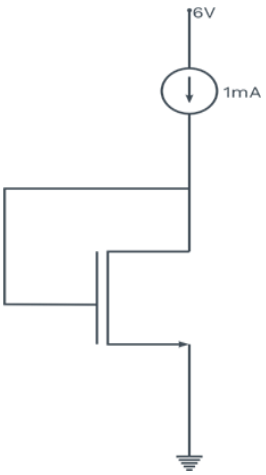
(C)Small signal gain of the amplifier is –6

(D)Small signal gain of the amplifier is –10

43. Consider the transistor in the circuit having following parameters:

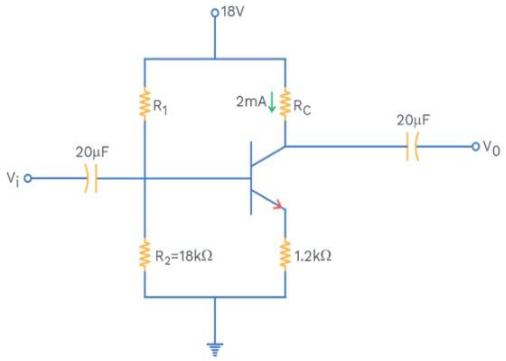
$V_{TN}$ = 1 V, $K_n$ = 1 $mAV^{-2}$ and $\lambda = 1V^{-1}$. The value of voltage $V_{DS}$ is _____V.

(Up to 2 decimal places).



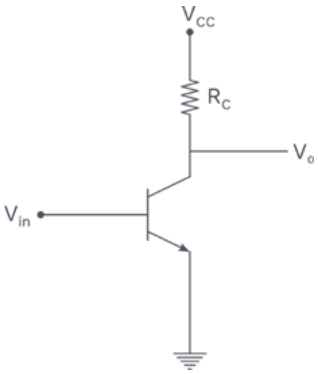
44. For the circuit shown below, the value of resistance $R_1$ is _____ $k\Omega$.

(up to two decimal places).




Assume $\beta$ is very large.

45. The transistor shown in figure below is biased with collector current of 2.6 mA and $R_C$ = 2.5 kΩ . If $\beta$ = 99 and $V_A$ = 13V, then small signal voltage gain is-



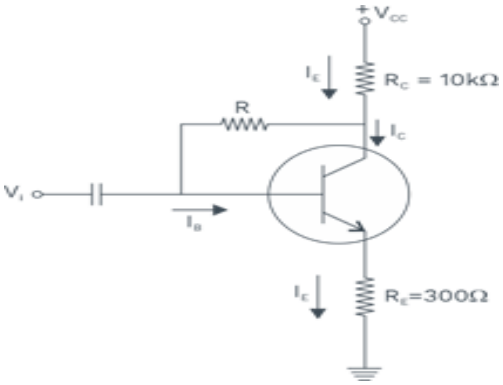

(A)–250 (B)–500

(C)–200 (D)–166.67

46. For the circuit shown below, find the value of R, if a silicon transistor is used with β = 50.

Given, $V_{CE}$ = 5V, $V_{CC}$ = 30 V, $R_C$ = 10 kΩ , and $R_E$ = 300 Ω,

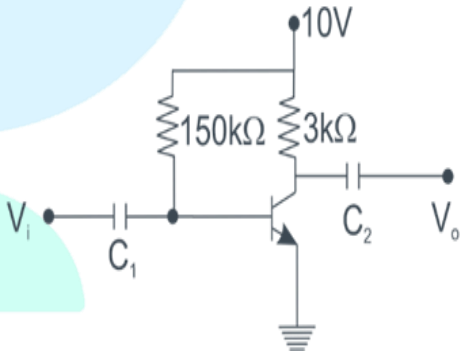
Neglect the reverse saturation current.



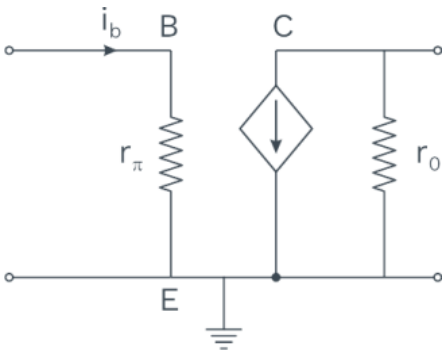

(A)107.2 kΩ (B)50 kΩ

(C)90.34 kΩ (D)30.23 kΩ

47. For the common-emitter configuration is shown below, the h-parameter is given by $h_{fe} = 50, h_{ie} = 0.5\text{k}\Omega, \text{h}_{oe} = 5\mu\text{S}$. The voltage gain is _____. (Up to 1 decimal place)



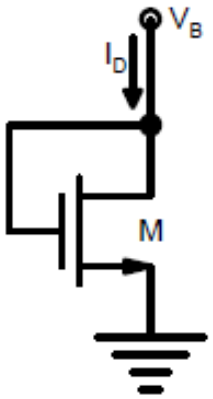

48. The current $i_b$ through the base of a silicon npn transistor is 1+0.1cos(10000πt)mA. At 300 K, the $r_\pi$ in the small signal model of the transistor is:



(A) $250\,\Omega$ (B) $27.5\,\Omega$

(C) $25\,\Omega$ (D) $22.5\,\Omega$

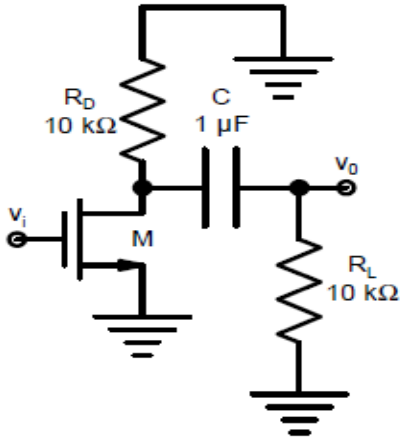
49. The small-signal resistance $\left(i.e.,dV_B/dI_D\right)$ in kΩ offered by the n-channel MOSFET M shown in the figure below, at a bias point of $V_B = 2V$, is (device data for M: device transconductance parameter $k_N = \mu_n C'_{ox}\left(W/L\right) = 40\mu A/V^2$, threshold voltage $V_{TN} = 1V$, and neglect body effect and channel length modulation effects)




(A) 12.5 (B) 25

(C) 50 (D) 100

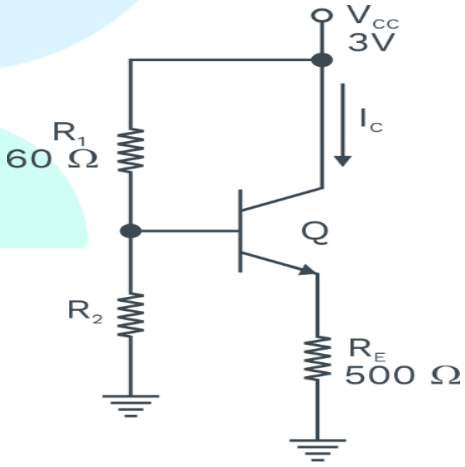
50. The ac schematic of an NMOS common-source stage is shown in the figure below, where part of the biasing circuits has been omitted for simplicity. For the *n*-channel MOSFET M, the transconductance $g_m = 1m\,A/V$, and body effect and channel length modulation effect are to be neglected. The lower cutoff frequency in Hz of the circuit is approximately at




(A) 8 (B) 32

(C) 50 (D) 200

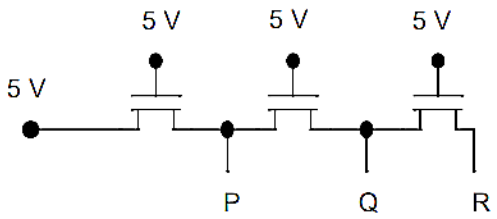
51. In the circuit shown below, the silicon npn transistor Q has a very high value of β. The required value of $R_2$ in kΩ to produce $I_C$ = 1 mA is




(A) 20 (B) 30

(C) 40 (D) 50

52. In the following circuit employing pass transistor logic, all NMOS transistors are identical with a threshold voltage of 1 V. Ignoring the body-effect, the output voltages at P, Q and R are



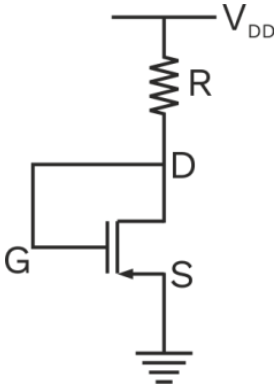

(A) 4 V, 3 V, 2 V

(B) 5 V, 5 V, 5 V

(C) 4 V, 4 V, 4 V

(D) 5 V, 4 V, 3 V

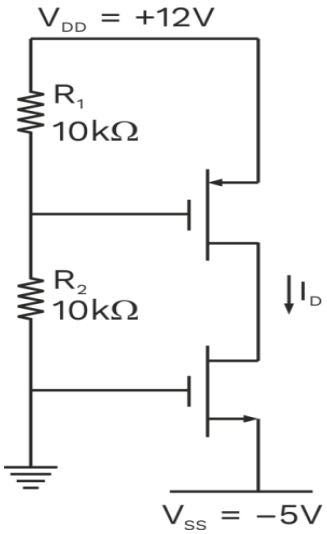
53. For the n-channel MOS transistor shown in the figure, the threshold voltage $V_{Th}$ is 0.8 V. Neglect channel length modulation effects. When the drain voltage $V_D$ = 1.6V. the drain current $I_D$ was found to be 0.5 mA. If $V_D$ is adjusted to be 2V by changing the values of R and $V_{DD}$, the new value of $I_D$ (in mA) is



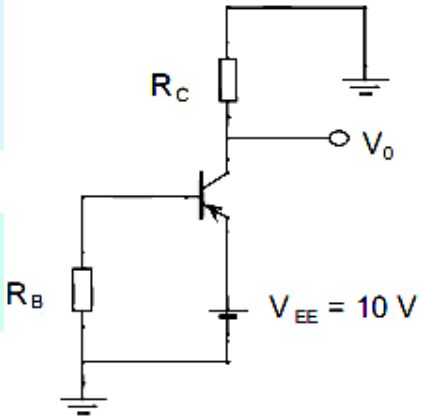

(A) 0.625 (B) 0.75

(C) 1.125 (D) 1.5

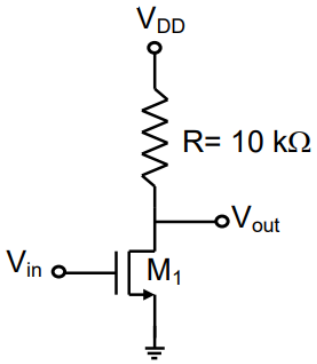
54. For the MOSFETs shown in the figure, the threshold voltage $|V_t| = 2V$ and $K = \frac{1}{2}\mu C_{ox}\left(\frac{W}{L}\right) 0.1 mA/V^2$. The value of $I_D$ (in mA) is _____.



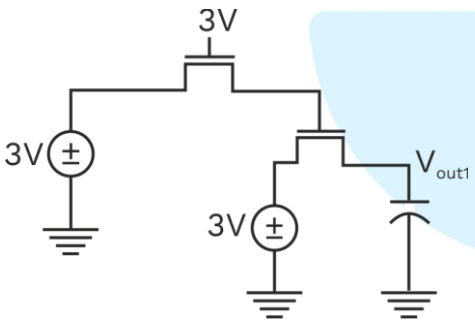

55. In the circuit shown, the PNP transistor has $|V_{BE}|$ = 0.7 Vand β = 50. Assume that $R_B$ = 100 $k\Omega$. For $V_0$ to be 5 V, the value of $R_C$ (in $k\Omega$) is_____.

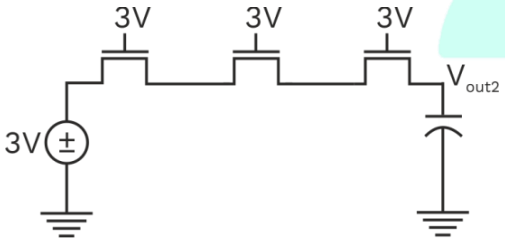



56. For the MOSFET $M_1$ shown in the figure, assume W/L = 2, $V_{DD}$ = 2.0 V, $\mu_n C_{ox}$ = 100 μA/$V^2$ and $V_{TH}$ = 0.5 V. The transistor $M_1$ switches from saturation region to linear region when Vin (in Volts) is_____.




57. In the circuits shown, the threshold voltage of each NMOS transistor is 0.6V. Ignoring the effect of channel length modulation and body bias, the values of $V_{out1}$ and $V_{out2}$, respectively, in volts, are:



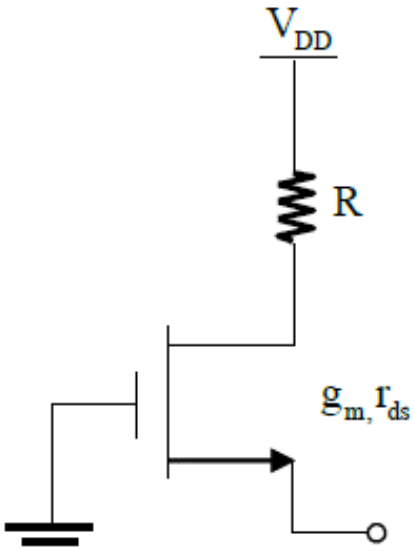

(A) 2.4 and 1.2

(B) 2.4 and 2.4

(C) 1.8 and 1.2

(D) 1.8 and 2.4

58. Using the incremental low frequency small-signal model of the MOS device, the Norton equivalent resistance of the following circuit is:



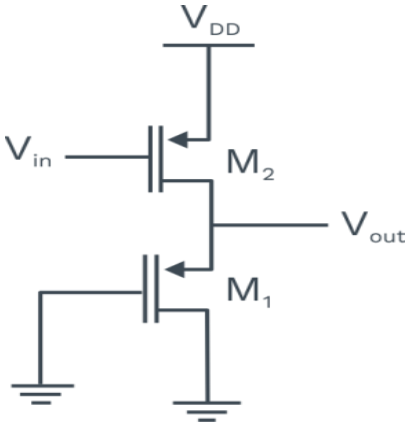

(A) $r_{ds} + R + g_m r_{ds} R$

(B) $\frac{r_{ds} + R}{1 + g_m r_{ds}}$

(C) $r_{ds} + \frac{1}{g_m} + R$

(D) $r_{ds} + R$

59. In the circuit shown in the figure, the transistors $M_1$ and $M_2$ are operating in saturation. The channel length modulation coefficients of both the transistors are non-zero. The transconductance of the MOSFETs $M_1$ and $M_2$ are $g_{m1}$ and $g_{m2}$, respectively and the internal resistance of the MOSFETs $M_1$ and $M_2$ are $r_{01}$ and $r_{02}$, respectively




Ignoring the body effect, the ac small signal voltage gain $(\partial V_{out} / \partial V_{in})$ of the circuit is

(A) $-g_{m2}\,(r_{01} \| r_{02})$

(B) $-g_{m2}\left(\frac{1}{g_{m1}} \| r_{02}\right)$

(C) $-g_{m1}\left(\frac{1}{g_{m2}} \| r_{01} \| r_{02}\right)$

(D) $-g_{m2}\left(\frac{1}{g_{m1}} \| r_{01} \| r_{02}\right)$

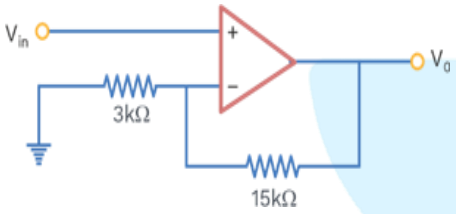
# Answer Key

Q.1 Ans. [ 4 ]
Q.2 Ans. [ C ]
Q.3 Ans. [ B ]
Q.4 Ans. [ 1.362 ]
Q.5 Ans. [ 2043.14 ]
Q.6 Ans. [ A ]
Q.7 Ans. [B ,D]
Q.8 Ans. [ B ]
Q.9 Ans. [ A ,B ]
Q.10 Ans. [ B ,D ]
Q.11 Ans. [6.07]
Q.12 Ans. [ B ]
Q.13 Ans. [A ,C ,D]
Q.14 Ans. [ 0.615 ]
Q.15 Ans. [ 3.92 ]
Q.16 Ans. [ D ]
Q.17 Ans. [ 158.73 ]
Q.18 Ans. [ A ,B ]
Q.19 Ans. [B]
Q.20 Ans. [ 48 ]
Q.21 Ans. [ B ]
Q.22 Ans. [ D ]
Q.23 Ans. [ C ]
Q.24 Ans. [A]
Q.25 Ans. [A, B, D]
Q.26 Ans. [ B ]
Q.27 Ans. [3.4]
Q.28 Ans. [ B ]
Q.29 Ans. [B]
Q.30 Ans. [ B ]
Q.31 Ans. [A]
Q.32 Ans. [ D ]
Q.33 Ans. [ B ]
Q.34 Ans. [ A ]
Q.35 Ans. [D]
Q.36 Ans. [ C ]
Q.37 Ans. [ C ]
Q.38 Ans. [A]
Q.39 Ans. [A ,C]
Q.40 Ans. [ B ,C ]
Q.41 Ans. [ D ]
Q.42 Ans. [A ,D]
Q.43 Ans. [ 1.62]
Q.44 Ans. [ 86.52 ]
Q.45 Ans. [ D ]
Q.46 Ans. [ C ]
Q.47 Ans. [ -295.6 ]
Q.48 Ans.[C]
Q.49 Ans.[A]
Q.50 Ans.[A]
Q.51 Ans.[C]
Q.52 Ans.[C]
Q.53 Ans.[C]
Q.54 Ans.[0.9]
Q.55 Ans. [0.75, Range 1.04 to 1.12]
Q.56 Ans. [1.5, Range 1.4 to 1.6]
Q.57 Ans.[D]
Q.58 Ans.[B]
Q.59 Ans. [D]

# 3 Operational Amplifier and its Applications

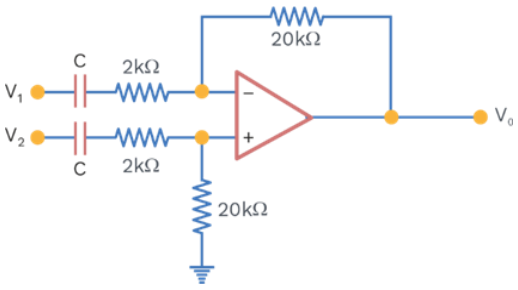
## MCQ, MSQ and NAT

1. The circuit containing an op-amp is shown in the figure below.




The op-amp has poor open-loop voltage gain which is given to be 48. Except for this case, the op-amp shows ideal characteristics. The gain of the op-amp is _____. (Up to 3 decimal places)

2. A circuit is shown in the figure below.




The input offset voltage of the op-amp is 4 mV. The output DC error voltage is -

(A)0 mV

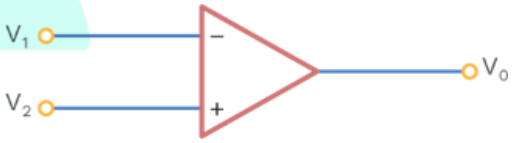
(B)2 mV

(C)4 mV

(D)40 mV

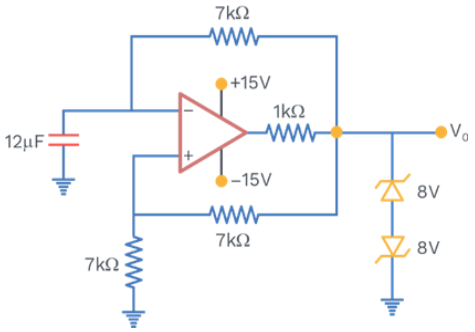
3. An RC differentiator circuit with R = 2 kΩ and C = 3 μF is applied with an input. The input to the circuit is a ramp voltage defined as v(t) = 250t Volts. The magnitude $V_0$ is _____ V. (Up to one decimal place)

4. A differential amplifier is shown in the figure below.




The differential mode gain is 100 and CMRR is 40dB. If the inputs to the amplifier are given as $V_1$ = 0.42 V and $V_2$ = 0.58 V, then the output voltage $V_0$ is_____ V. (Up to one decimal place)

5. A circuit is shown in shown on the figure below.




The supply voltage is ±15V. The output waveform has-

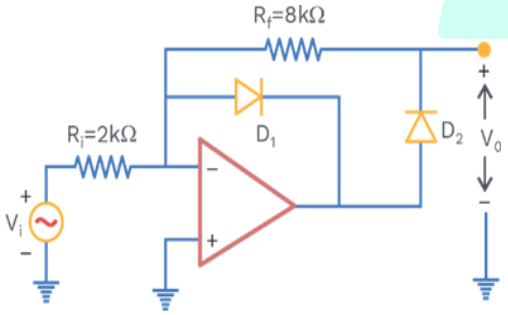
(A)an amplitude of ± 15V and a time period of 58.22 msec.

(B)an amplitude of ± 8V and a time period of 116.44 msec.

(C)an amplitude of ± 8V and a time period of 58.22 msec.

(D)an amplitude of ± 15V and a time period of 116.44 msec.

6. For the circuit shown in the figure below, which of the following statements is/are TRUE?




Consider the cut-in voltage of the diode as 0.7 V.

(A)When $V_i > 0$; $V_0 = \frac{-R_f}{R_i} V_i$

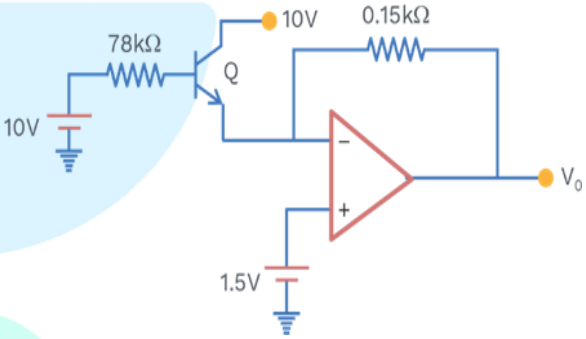
(B)When $V_i < 0$; $V_0 = -4V_i$

(C)When $V_i < 0$; $V_0 = 0V$

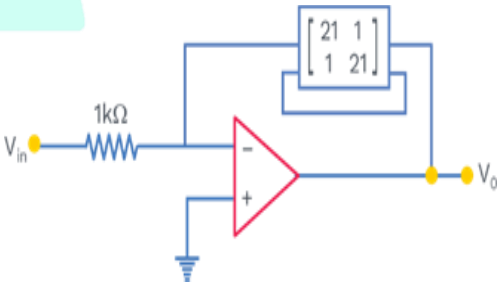
(D)When $V_i > 0$; $V_0 = 0V$

7. The supply voltage($V_{cc}$) to a non inverting op-amp is 10 V. The op-amp gain is 22. The slew rate of the op-amp is 0.42 × $10^6$ V/s. For a sinusoidal input of amplitude 0.5V, the frequency for distortionless operation is _____ kHz. (Up to 3 decimal places)

8. For the circuit shown in the figure below, it is given that $V_{BE}$ = 0.7 V and β = 99 for the transistor. The output voltage $V_0$ is _____ V. (Rounded off to the nearest integer).



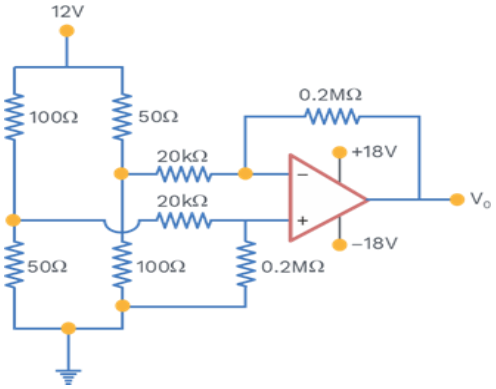

9. Consider the circuit shown below.



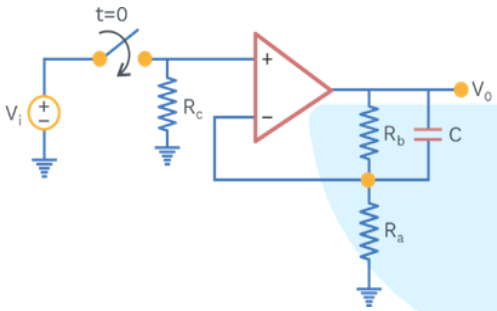

The parameters connected in the feedback are Z-parameters ( in$k\Omega$). The value of $\left|\frac{V_0}{V_{in}}\right|$ is _____. (Rounded off to the nearest integer)

10. The output voltage for the circuit shown in the figure below is _____V.

(Rounded off to the nearest integer)

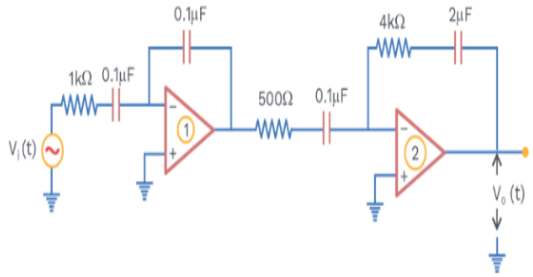



11. An op-amp circuit is shown in the figure below.




$V_i$ = 2V, $R_a$ = 10 kΩ, $R_b$ = 42 kΩ, $R_C$ = 10 kΩ and C = 1 μF. The switch is closed at t = 0. The output voltage at time t = 0.21 msec is _____ V. (Up to two decimal places)

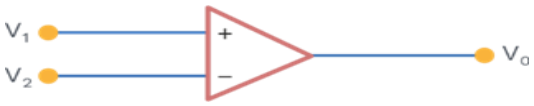
12. A signal $V_i(t)$ = 2 + 2 sin (100πt) + 2 sin (5000πt) + 2 sin (10000πt) is applied to the filter circuit. The least attenuated frequency in the output is-




(A)100 Hz (B)7.29 kHz

(C)13.13 kHz (D)102.1 kHz

13. A circuit is shown in the figure below.




If the differential gain is 100000 and CMRR is 1000, then which of the following statement(s) is/are true?

[Consider $V_1 = 400\mu V$ and $V_2 = 260\mu V$]

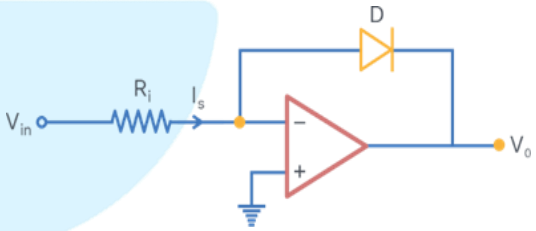
(A)Open loop output, $V_{OL}$ = 14.33 V

(B)Open loop output, $V_{OL}$ = 14.033V

(C)Common mode output, $V_{ocm}$ = 33 mV

(D)Common mode gain, $A_{cm}$ = 10

14. A circuit is shown in the figure below.




When, $V_{in}$ = 3V, $V_0 = V_{01}$ and when $V_{in}$ = 6V, $V_0 = V_{02}$. The relation between $V_{01}$ and $V_{02}$ is-

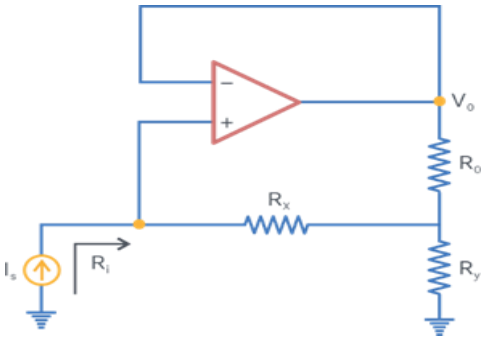
(A)$V_{0_1} - V_{0_2} = \ln(3)$

(B)$V_{0_1} - V_{0_2} = V_T \ln(2)$

(C)$V_{0_1} - V_{0_2} = V_T \ln\left(\frac{3}{R_i I_s}\right)$

(D)$V_{0_1} - V_{0_2} = -V_T \ln(2)$

15. A circuit is shown in the figure below.

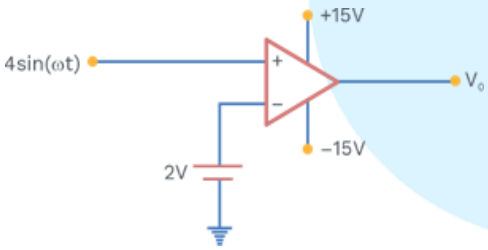



The value of '$R_i$' is _____ $k\Omega$. (Rounded off to the nearest integer).

Consider

$R_0 = 6\ k\Omega, R_x = 30\ k\Omega, R_y = 2\ k\Omega$.

16. Consider the circuit shown in the figure below.




The duty cycle (in %) is-

(A)50

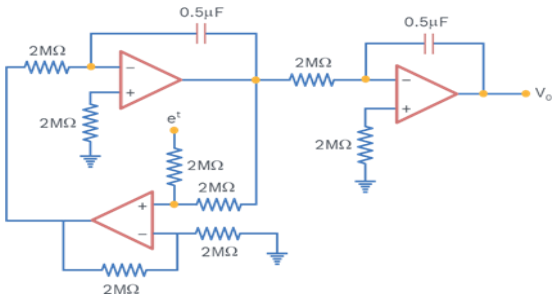
(B)33.33

(C)16.66

(D)12.5

17. The output voltage of the given circuit is expressed as-

$$\frac{d^2V_0}{dt^2} + A\frac{dV_0}{dt} + BV_0 = x(t)$$




The value of x(t) and B is -

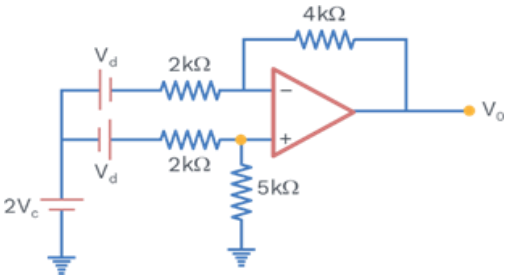
(A) $x(t) = e^{-t}$, B = 1

(B) x(t) = B = 0

(C) $x(t) = e^{t}$, B = 0

(D) x(t) = 0, B = 1

18. A circuit is shown in the figure below.




Where

$V_d$ = Differential voltage

$V_c$ = Common-mode voltage

Which of the following statement(s) is/are is true?

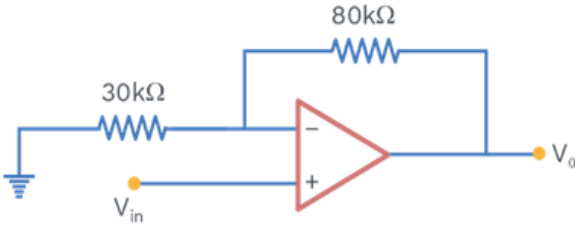
(A) Common-mode gain = $\frac{29}{7}$

(B) Differential gain = $\frac{29}{7}$

(C) CMRR = 14.5 dB

(D) CMRR = 23.22 dB

19. The OP-AMP circuit shown below has an open-loop voltage gain of 200. Find the gain of the amplifier if the OP-AMP is ideal.

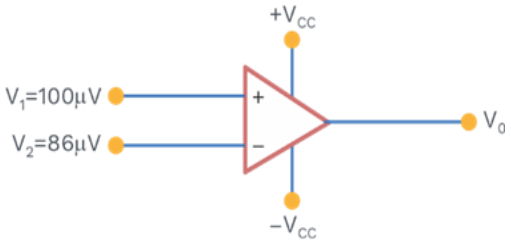



(A)3.66 (B)0.27

(C)20 (D)25.6

20. The differential operational

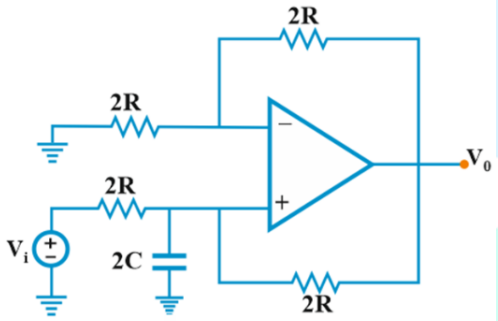
amplifier circuit is given below. The op Amp has a common-mode gain of 200, differential mode gain of $10^4$, and CMRR of 500. The percentage error in the output $V_0$ is -




(A)3.25% (B)2.68%

(C)1.33% (D)0.92%

21. For the given op-amp circuit, $\frac{V_0(s)}{V_i(s)}$ is -

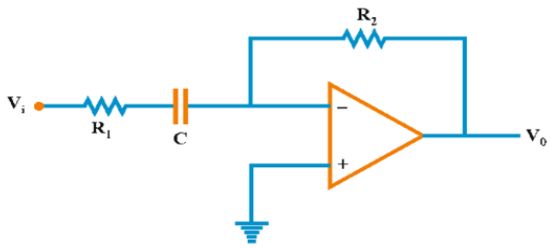



(A)$\frac{1}{sCR}$ (B)$\frac{1}{2sCR}$

(C)$\frac{1}{sC}$ (D)$\frac{1}{2sC}$

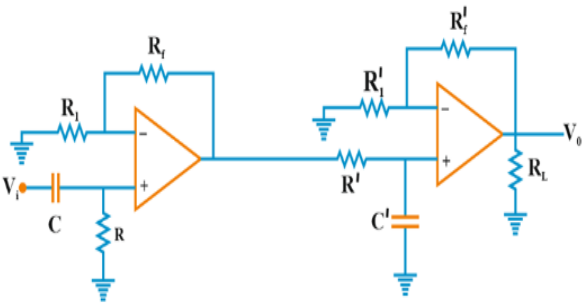
22. An op-amp has non-inverting input of 745 μV and inverting input of 740 μV. The differential mode gain of the op-amp is 5 × $10^5$ and its CMRR is 80 dB. The percentage error due to the common mode is _____%. (Up to two decimal places)

23. A high pass filter circuit is shown in the figure below




If $R_1$ = 100 kΩ , $R_2$ = 1 MΩ and 3–dB frequency is 1 kHz, then the value of capacitor C is _____pF. (Up to one decimal place)

24. Consider a bandpass filter circuit as shown in the figure below.




Given that

$R = R' = 10k\Omega, C = 0.1\mu F, C' = 0.0025\mu F$

Which of the following statement(s) is/are correct?

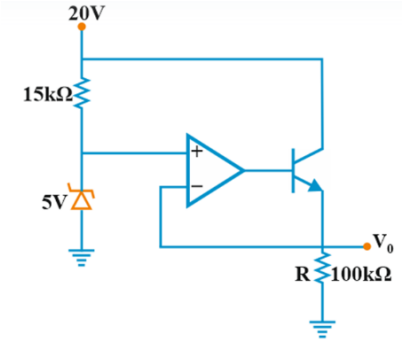
(A)Higher cut-off frequency, $f_H$ = 6.36 kHz

(B)Lower cut-off frequency, $f_L$ = 159.15 Hz

(C)Bandwidth, BW = 6.2 kHz

(D)Center frequency, $f_c$ = 1 Hz

25. An op-amp based circuit is shown below.



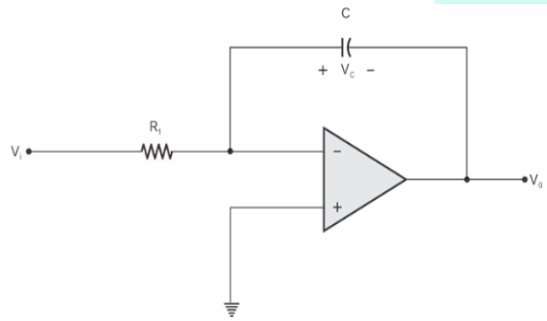

Assume op-amp is ideal and β to be very large. The total current due to 20 V supply voltage is –

(A)40 mA (B)1.05 mA

(C)45 mA (D)5 mA

26. A differential amplifier is having a differential voltage gain of 26 dB and the common mode rejection ratio of 23 dB. The common mode voltage gain will be _____dB.[Rounded off to the nearest integer]

27. For the op-amp circuit shown below, the output voltage $V_0$ is given by-



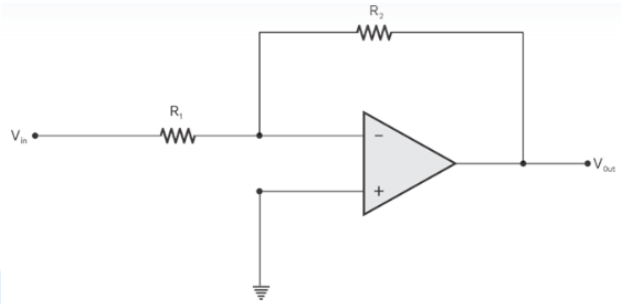

(A) $V_0 = -\frac{1}{R_1C}\int V_i dt$

(B) $V_0 = -R_1C\int V_i dt$

(C) $V_0 = -R_1C\frac{dV_i}{dt}$

(D) $V_0 = -\frac{1}{R_1C}\frac{dV_i}{dt}$

28. The given amplifier circuit is designed to provide a voltage gain of '-200 V/V' and an input resistance of 2 k Ω as seen by the source $V_{in}$. The probable values of $R_1$ and $R_2$ will be respectively-




(A)2 kΩ and 200 kΩ

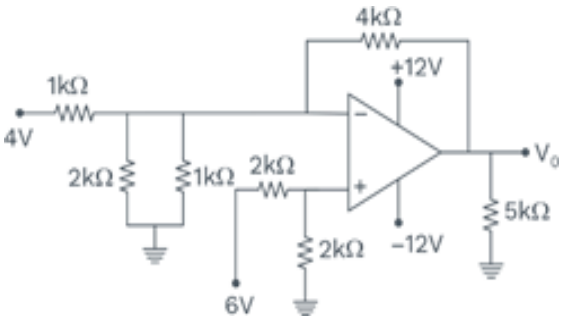
(B)1 kΩ and 200 kΩ

(C)2 kΩ and 400 kΩ

(D)1 kΩ and 400 kΩ

29. A 20 mV input to an amplifier gives an output of 20 V without feedback. When a negative feedback is employed, 200 mV input is required to obtain the same output. The feedback factor ($\beta$) will be_____%.[Up to 1 decimal place].

30. Consider the op-amp circuit shown below:

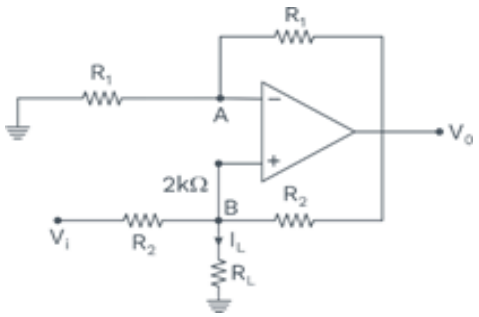


If the op-amp is ideal, then the value of output voltage $V_0$ is equal to.

(A)8 V (B)17 V

(C)12 V (D)10 V

31. Consider the op-amp circuit shown below:




If the op-amp is ideal, find the current through the load resistance $R_L$.

(A) $I_L = \frac{V_i}{R_2 + R_L}$

(B) $I_L = \frac{V_i}{R_L}$

(C)$I_L = 0$

(D) $I_L = \frac{V_i}{R_2}$

32. An amplifier has an open loop gain of 100, an input impedance of 5kΩ, and an output impedance of 200Ω. A feedback network with a feedback factor of 0.99 is connected to the amplifier in a voltage series feedback mode. The new input and output impedances, respectively, are:

(A)50Ω and 2Ω

(B)50Ω and 20Ω

(C)500kΩ and 2Ω

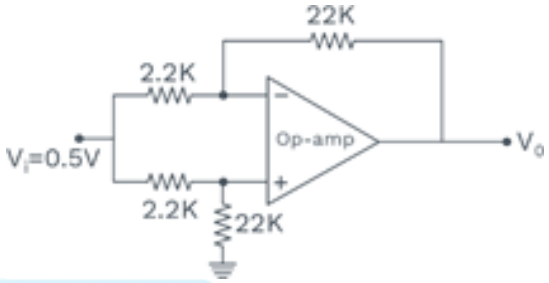
(D)500kΩ and 1Ω

33. For a given op-amp, CMRR = $10^5$ and differential gain = $10^8$. What is the common mode gain of the op-amp?

(A)$10^3$ (B)$2\times10^5$

(C)$10^5$ (D) 1

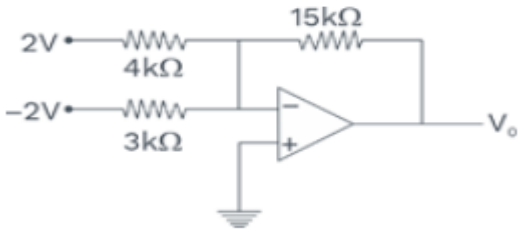
34. In the op-amp circuit shown below (assuming ideal op-amp), find the output voltage $V_0$.




(A)$V_0 = -5V$ (B)$V_0 = 5V$

(C)$V_0 = 0$ (D)$V_0 = -2V$

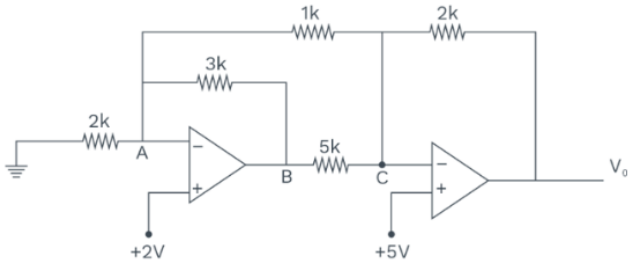
35. An ideal operational amplifier circuit is shown below. Find the output voltage of the operational amplifier in volts?




(A)0 (B)1.5

(C)2.5 (D)3.5

36. The output voltage $(V_0)$ for the circuit is



(A)14 volts (B)5 volts

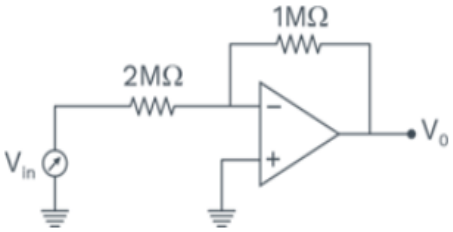
(C)0 volts (D)10 volts

37. An op-amp has a gain-bandwidth product of 5 MHz. A non-inverting amplifier using this op-amp and having a voltage gain of 40 dB will exhibit a 3 dB bandwidth of -

(A)25 kHz (B)50 kHz

(C)100 kHz (D)33.3 kHz

38. The given op-amp has an offset voltage of 2mV. The output voltage for the circuit shown in the figure below is:




(A)1 mV (B) 0.5V

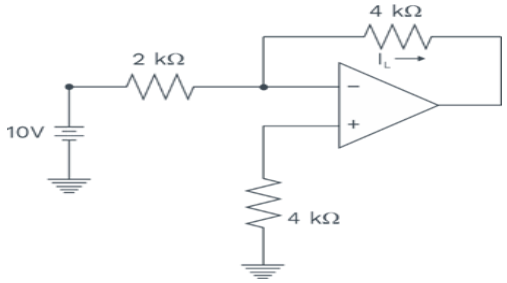
(C)1 V (D)± 1V

39. A differential amplifier is given whose differential gain is 10000 and its CMRR is given to be 80 dB. The common mode gain is

(A)2 (B)1

(C)$\frac{1}{2}$ (D)0

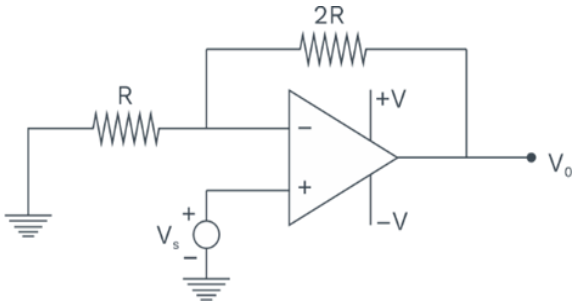
40. Determine $I_L$ for the following circuit-




(A)5 mA (B)4 mA

(C)2 mA (D)6 mA

41. For the operational amplifier circuit shown below, the parameter (A) is determined by $\frac{V_0}{V_s}$.




The value of A is

(A)-2 (B)2

(C)3 (D)-3

42. For an ideal differential amplifier, the common mode rejection ratio (CMMR) should be-

(A)As high as possible

(B)As low as possible

(C)Constant

(D)None of these

43. An op-amp has an open-loop gain of $10^5$ and an open loop upper cut-off frequency of 10 Hz. If this op-amp is connected as an amplifier with a closed loop gain of 100, then the upper cut-off frequency is-

(A)10 Hz (B)100 Hz

(C)10 kHz (D)100 kHz

44. The gain stability of an amplifier can be improved by using-

(A)positive feedback

(B) negative feedback

(C)both positive and negative feedback

(D)none of these

45. A negative feedback amplifier where an input current controls an output voltage is called-

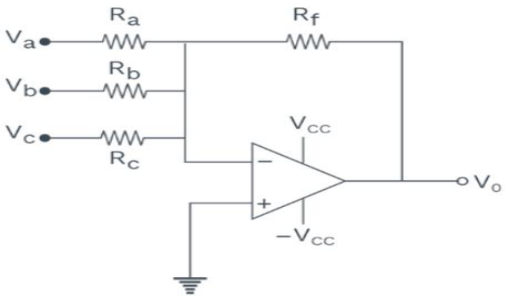
(A)Current amplifier

(B)Trans-conductance amplifier

(C)Trans-resistance amplifier

(D)Trans-resistance amplifier

46. In the summing amplifier shown below




$V_a = 1V, V_b = 2V, V_c = 3\ \mathrm{V}$

$R_a = R_b = R_c = 3\mathrm{k}\Omega, R_f = 1\mathrm{k}\Omega$

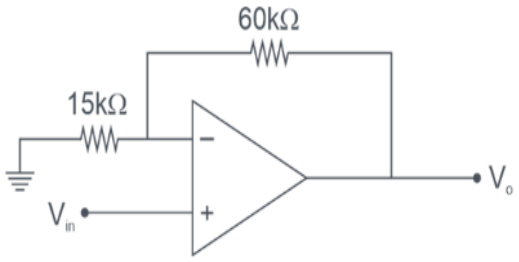
Supply voltage = ±15V

Determine the output voltage $V_0$.

(A)2 V (B)-2 V

(C)4 V (D)-4 V

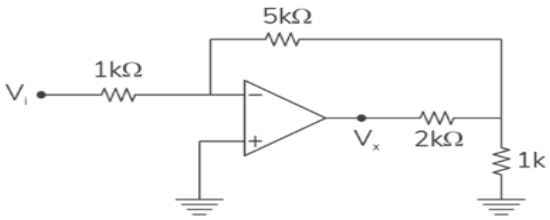
47. An Op-amp circuit shown below has open loop voltage gain of 100, find the gain of the amplifier if Op-amp is ideal -




(A)0.2 (B)5

(C)20 (D)5

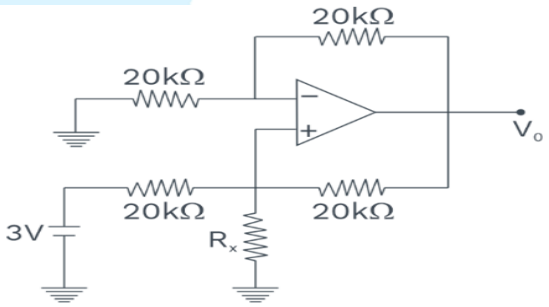
48. Consider an ideal amp as shown in figure. Find the voltage $\mathrm{V_x}$ ?




(A)$-7V_i$ (B)$17V_i$

(C)$-17V_i$ (D)$7\ \mathrm{V_i}$

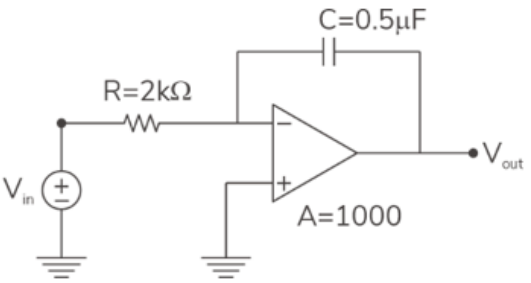
49. The output voltage of the given operational amplifier current is 10V, what is the value of $R_x$ as shown in the circuit?




(A)$13.33\mathrm{k}\Omega$ (B)$23.33\mathrm{k}\Omega$

(C)$33.33\mathrm{k}\Omega$ (D)$66.66\mathrm{k}\Omega$

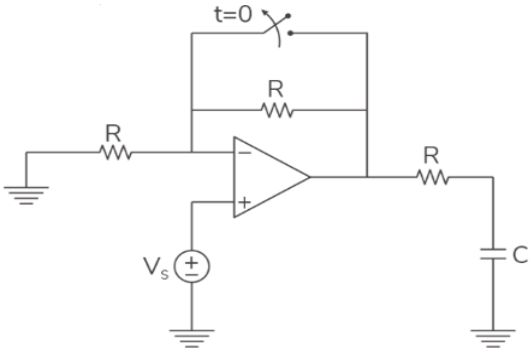
50. Consider an op-amp circuit shown below having open loop gain A = 1000. Then the time constant of the circuit is _____ (m sec).



(A)1000 (B)1001

(C)1 (D)1001

51. Consider an op-amp circuit shown, switch opens at t = 0 and for t < 0, circuit is in steady state. Then $V_C(t)$ for $t > 0$?



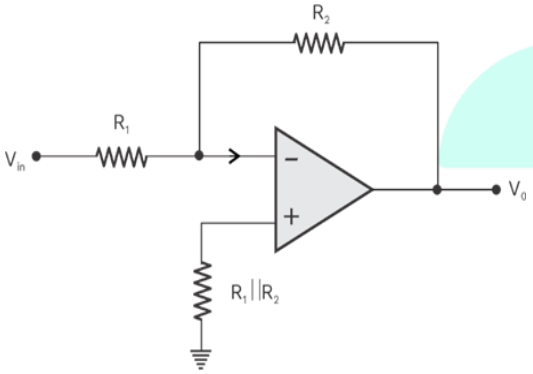

(A)$V_S + V_S e^{-t/RC}$

(B)$V_S + V_S e^{-tRC}$

(C)$2V_S - V_S e^{-t/RC}$

(D)$2V_S + V_S e^{-tRC}$

52. An op-amp circuit is shown below




Given,

$R_1$ = 2 kΩ

$R_2$ = 200 kΩ

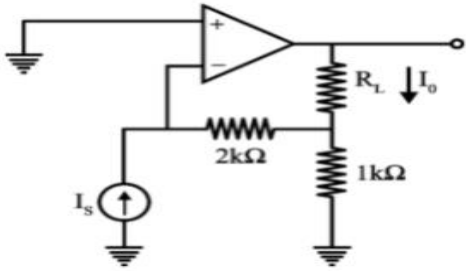
Input offset voltage drift = 7 µV/°C

Input offset current drift = $\frac{1}{2}\text{nA}/^\circ\text{C}$

The amplifier circuit voltage = 0 at 30°C for input voltage. If the input voltage is 11mV, the value of output voltage at 50°C is_____ V. (Upto 3 decimal places)

[Consider the input offset voltage at non-inverting terminal as positive].

53. Consider the circuit as shown in the figure below.




Which of the following statement(s) is/are correct?

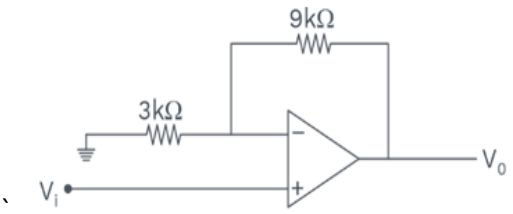
(A)Feedback type is shunt-series.

(B)Feedback factor, $\beta = \frac{1}{2}$

(C)Feedback type is series - shunt.

(D)Feedback factor, $\beta = \frac{1}{3}$

54. A circuit containing op-amp is shown in the figure below.

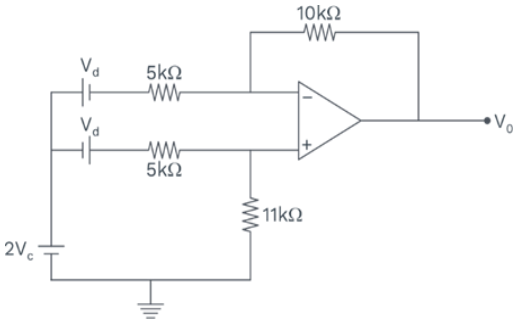



The op-amp open loop gain is 60 but is ideal otherwise. The gain of the amplifier is-

(A)2.5 (B)3.75

(C)4 (D)5.5

55. A circuit is shown in the figure below.




The output of the circuit ($V_0$) is given as-

$V_0 = A_dV_d + A_cV_c$

where,

$A_d$ = Differential gain

$A_c$ = Common mode gain

Which of the following statement(s) is/are TRUE?

(A)If $V_d = 0$, then $A_c = \frac{1}{16}$

(B)If $V_d = 0$, then $A_c$ = -18.061 dB

(C)Differential gain is equal to $\frac{65}{16}$

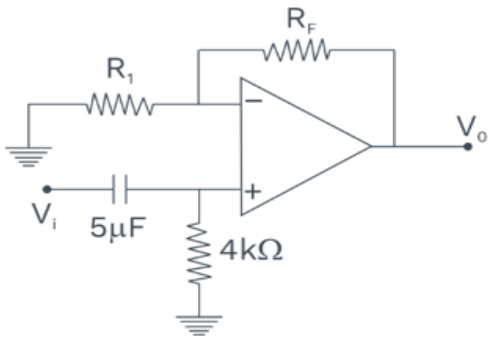
(D)CMRR is equal to 30.23 dB

56. An inverting amplifier having a gain of 100 has a slew rate of 0.5 V/$\mu$sec. What is the maximum input amplitude applied to it, if the input signal has a frequency of 18 kHz?

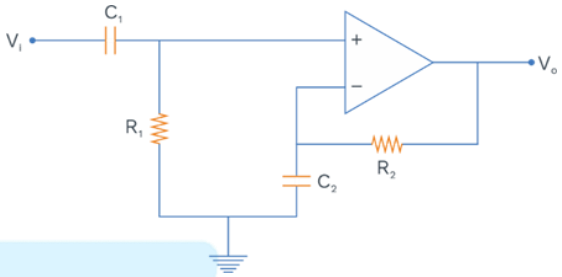
(A)56 mV (B)54 mV

(C)44.19 mV (D)34 mV

57. For the operational amplifier shown with filter circuit, the cut-off frequency of the filter is _____rad/sec.




58. The op-amp circuit is shown below.




If the value of resistance and capacitance is same, then the op-amp is a

(A)Low pass filter with pole at origin

(B)High pass filter with pole at origin

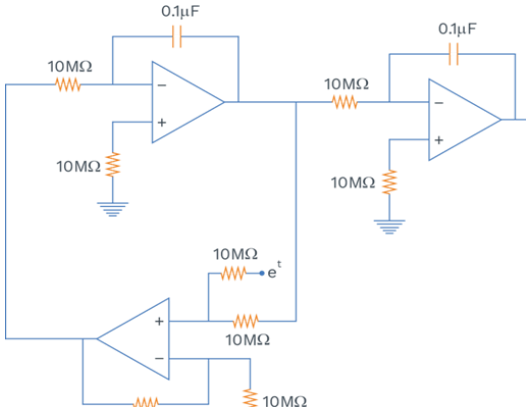
(C)Low pass filter with pole at $-\frac{1}{RC}$

(D)High pass filter with pole at $-\frac{1}{RC}$

59. The circuit produces an output $V_0$ which is given in the form of-

$$\frac{d^2 V_0}{dt^2} + \alpha\frac{dV_0}{dt} + \beta V_0 = x(t)$$

The circuit is given as

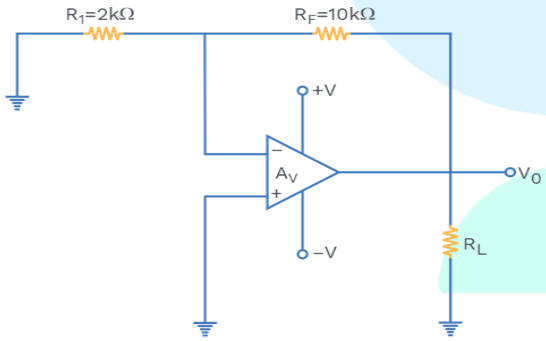



The value of $\alpha$ and $\beta$ are respectively-

(A)–1 and 1 (B)0 and 1

(C)1 and 0 (D)–1 and 0

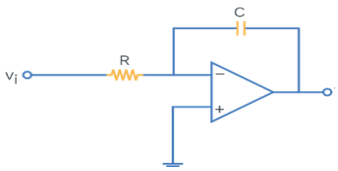
60. In the circuit shown below, the maximum output voltage $V_{0max}$ for op amp $V_{io}$ = 6mV (maximum), $I_{io}$ = 50 nA (maximum), $I_B$ = 250 nA (maximum) is



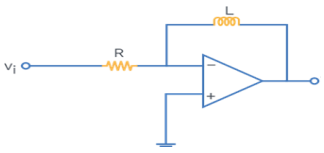

(A)0 mV (B)20 mV

(C)38.5 mV (D)8.3 mV

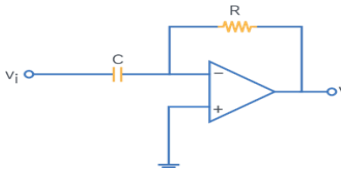
61. Which of the following circuit is/are a differentiator?



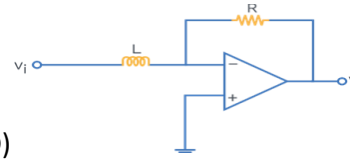

(A)




(B)




(C)



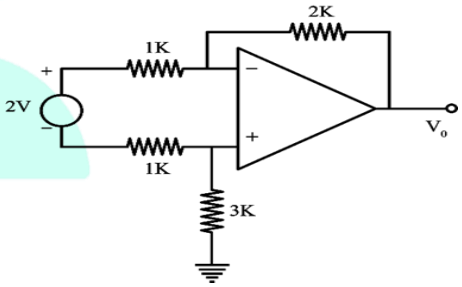

(D)

62. An amplifier using an op-amp with a slew rate of 1V/μ-sec has voltage gain of 40 dB. If this amplifier has to faithfully amplify a sinusoidal signal having frequency 50 kHz without any distortion, then the possible value of peak input voltage is

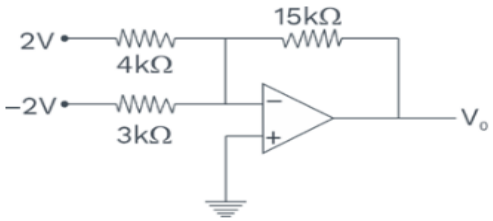
(A)32 mV (B)100 mV

(C)20 mV (D)50 mV

63. Find '$V_0$' in volts if op-amp is ideal?




64. An ideal operational amplifier circuit is shown below. Find the output voltage of the operational amplifier in volt.




(A)0 (B)1.5

(C)2.5 (D)3.5

65. Which of the following statements is/are correct regarding CMRR of an op amp?

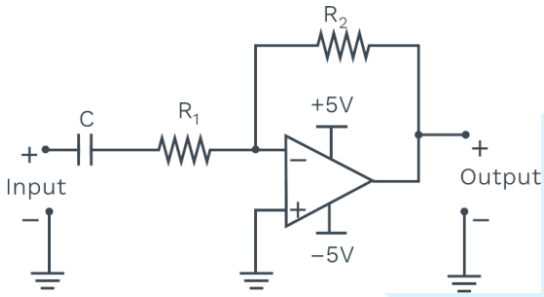
(A)It is the ability of the op-amp to reject unwanted signal.

(B)It should be ideally zero.

(C)It should be very high ideally.

(D)If CMRR is high, the common mode noise voltage would be low.

66. The circuit shown is a



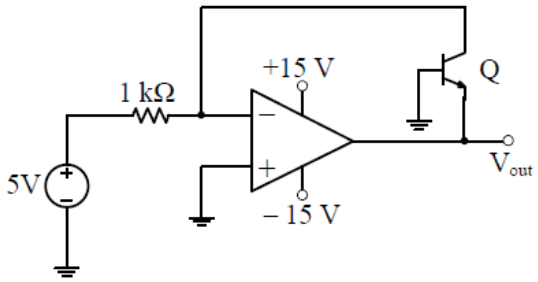

(A) low pass filter with $f_3dB = \frac{1}{(R_1 + R_2)C} rad / s$

(B) high pass filter with $f_3dB = \frac{1}{R_1 C} rad / s$

(C) low pass filter with $f_3dB = \frac{1}{R_1 C} rad / s$

(D) high pass filter with $f_3dB = \frac{1}{(R_1 + R_2)C} rad / s$

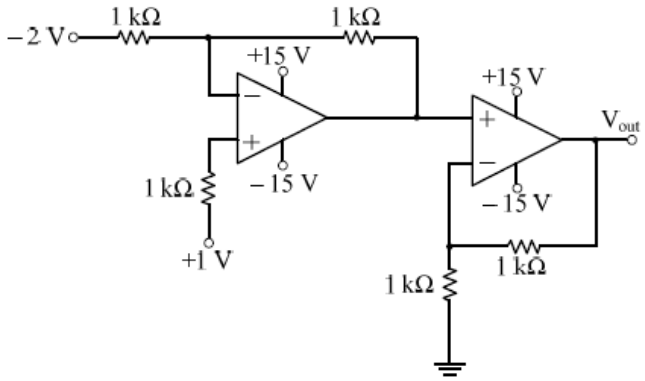
67. In the circuit shown below what is the output voltage ($V_{out}$) if a silicon transistor Q and an ideal op-amp are used?




(A) –15 V (B) –0.7 V

(C) +0.7 V (D) +15 V

68. In the circuit shown below the op amps are ideal. Then $V_{out}$ in Volts is




(A) 4 (B) 6

(C) 8 (D) 10

69. A good current buffer has

(A) low input impedance and low output impedance

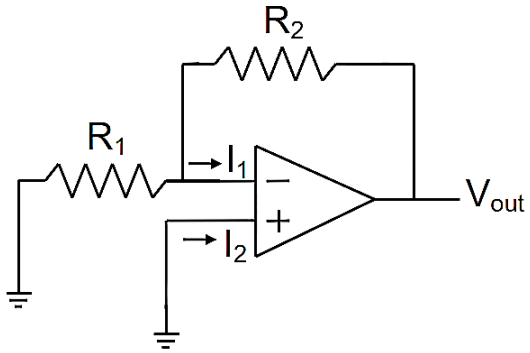
(B) low input impedance and high output impedance

(C) high input impedance and low output impedance

(D) high input impedance and high output impedance

70. In the circuit shown, the op-amp has finite input impedance, infinite voltage gain and zero input offset voltage. The

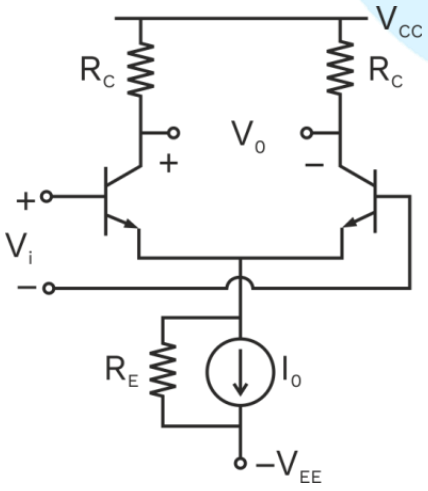
output voltage $V_{out}$is:




(A) $-I_2(R_1 + R_2)$ (B) $I_2R_2$

(C) $I_1R_2$ (D) $-I_1(R_1 + R_2)$

71. In the differential amplifier shown in the figure, the magnitudes of the common-mode and differential-mode gains are $A_{cm}$ and $A_d$, respectively. If the resistance $R_1$ is increased, then.



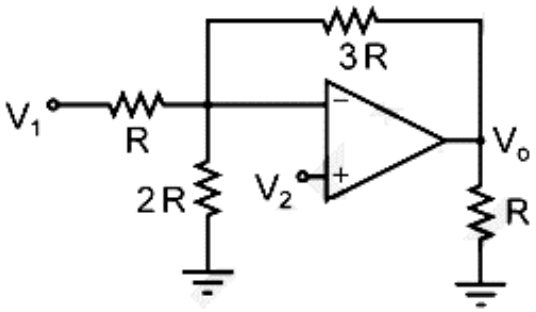

(A) $A_{cm}$ increases

(B) common-mode rejection ratio increases

(C) $A_d$ increases

(D) common-mode rejection ratio decreases

72. Assuming that the Op-amp in the circuit shown is ideal, $V_o$ is given by



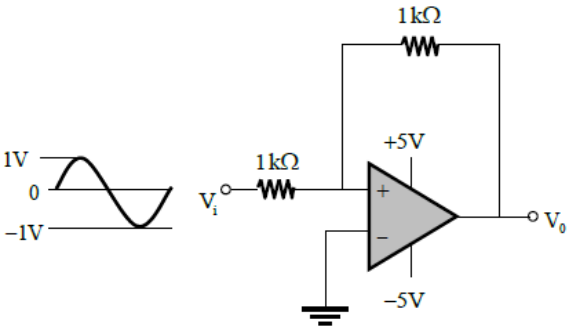

A. $\frac{5}{2}V_1$-$3V_2$

B. $2V_1$-$\frac{5}{2}V_2$

C. -$\frac{3}{2}V_1$+$\frac{7}{2}V_2$

D. -$3V_1$+$\frac{11}{2}V_2$

73. The components in the circuit shown below are ideal. If the op-amp is in positive feedback and the input voltage $V_i$ is a sine wave of amplitude 1V, the output voltage $V_0$ is



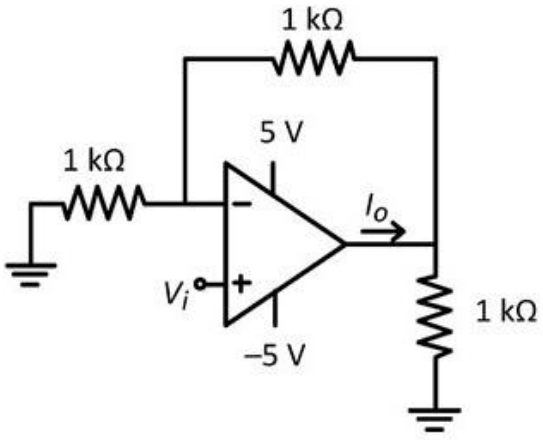

(A)a non-inverted sine wave of 2 V amplitude.

(B)an inverted sine wave of 1 V amplitude.

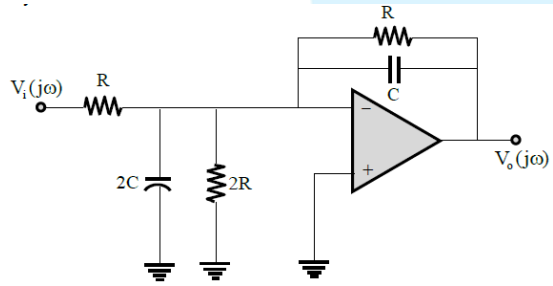
(C)a square wave 5 V amplitude.

(D)a constant of either + 5 V or -5 V.

74. In the circuit shown below, all the components are ideal. If $V_i$ is +2V, the current $I_0$ sourced by the op-amp is _____ mA.



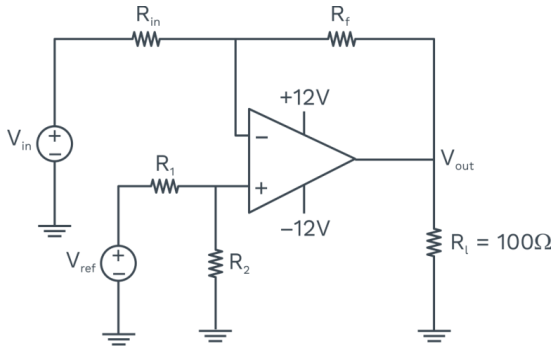

75. The components in the circuit given below are ideal. If R=2kΩ and C=1μF, the –3 dB cut-off frequency of the circuit in Hz is




(A) 14.92 (B) 34.46

(C) 59.68 (D) 79.58

76. For the circuit with an ideal Op-amp shown in the figure, $V_{ref}$ is fixed.

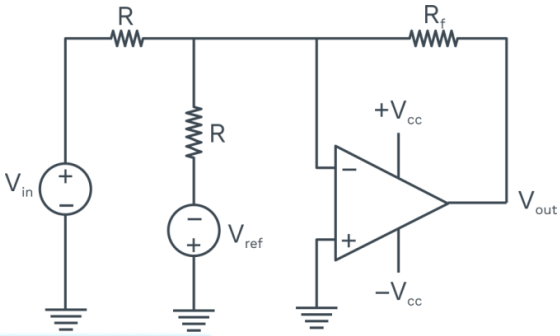



If $V_{out}$ = 1 volt for $V_{in}$ = 0.1 volt and $V_{out}$ = 6 volt for $V_{in}$ = 1 volt, where $V_{out}$ is measured across $R_L$ connected at the output of this Op-amp, the value of $R_f/R_{in}$ is

(A) 3.285 (B) 2.860

(C) 3.825 (D) 5.555

77. Consider the circuit with an ideal op amp shown in the figure.




Assuming

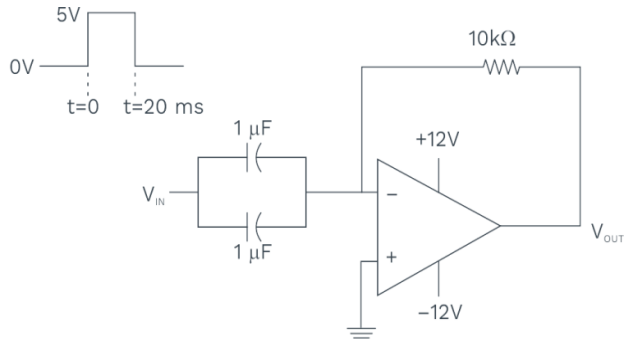
$$|V_{in}| << |V_{cc}| \text{ and } |V_{ref}| << |V_{cc}|,$$

the condition at which $V_{out}$ equals to zero is

(A) $V_{in} = V_{ref}$ (B) $V_{in} = 0.5\ V_{ref}$

(C) $V_{in} = 2\ V_{ref}$ (D) $V_{in} = 2 + V_{ref}$

78. A circuit with an ideal OPAMP is shown in the figure. A pulse $V_{IN}$ of 20 ms duration is applied to the input. The capacitors are initially uncharged.




The output voltage $V_{OUT}$ of this circuit at t = $0^+$ (in integer) is _____ V

## Answer Key

Q.1 Ans. [5.333]
Q.2 Ans. [C]
Q.3 Ans. [ 1.5]
Q.4 Ans. [16.5]
Q.5 Ans. [ B ]
Q.6 Ans. [ B ,D ]
Q.7 Ans. [6.076]
Q.8 Ans. [ 0 ]
Q.9 Ans. [440]
Q.10 Ans. [ -18 ]
Q.11 Ans. [ 2.04 ]
Q.12 Ans. [B]
Q.13 Ans. [ B ,C ]
Q.14 Ans. [ B ]
Q.15 Ans. [ 42 ]
Q.16 Ans. [ B ]
Q.17 Ans. [ C ]
Q.18 Ans. [ B ,D ]
Q.19 Ans. [ A ]
Q.20 Ans. [ C ]
Q.21 Ans. [ B ]
Q.22 Ans. [1.45]
Q.23 Ans. [112.8]
Q.24 Ans. [ A ,B ,C ]
Q.25 Ans. [ B ]
Q.26 Ans. [3]
Q.27 Ans. [A]
Q.28 Ans. [C]
Q.29 Ans. [0.9]
Q.30 Ans. [C]
Q.31 Ans. [D]
Q.32 Ans. [C]
Q.33 Ans. [ A ]
Q.34 Ans. [C]
Q.35 Ans. [C]
Q.36 Ans. [A]
Q.37 Ans. [B]
Q.38 Ans. [C]
Q.39 Ans. [B]
Q.40 Ans. [A]
Q.41 Ans. [C]
Q.42 Ans. [A]
Q.43 Ans. [C ]
Q.44 Ans. [B]
Q.45 Ans. [C]
Q.46 Ans. [ B ]
Q.47 Ans. [B]
Q.48 Ans. [C]
Q.49 Ans. [C]
Q.50 Ans. [B]
Q.51 Ans. [C]
Q.52 Ans. [ -1.083 ]
Q.53 Ans. [A ,D]
Q.54 Ans. [B]
Q.55 Ans. [ B ,C ,D ]
Q.56 Ans. [ C ]
Q.57 Ans. [50]
Q.58 Ans. [ B ]
Q.59 Ans. [C]
Q.60 Ans. [ C ]
Q.61 Ans. [ B ,C ]

**Q.62 Ans. [ C ]**

**Q.63 Ans. [ -5 ]**

**Q.64 Ans. [ C ]**

**Q.65 Ans. [ A ,C ,D ]**

**Q.66 Ans. [B]**

**Q.67 Ans. [B]**

**Q.68 Ans. [C]**

**Q.69 Ans. [B]**

**Q.70 Ans. [C]**

**Q.71 Ans. [B]**

**Q.72 Ans. [D]**

**Q.73 Ans. [D]**

**Q.74 Ans. [6]**

**Q.75 Ans. [D]**

**Q.76 Ans. [D]**

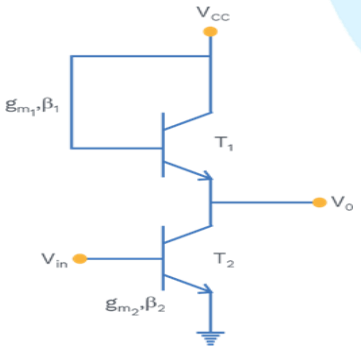
**Q.77 Ans. [A]**

**Q.78 Ans. [-12]**

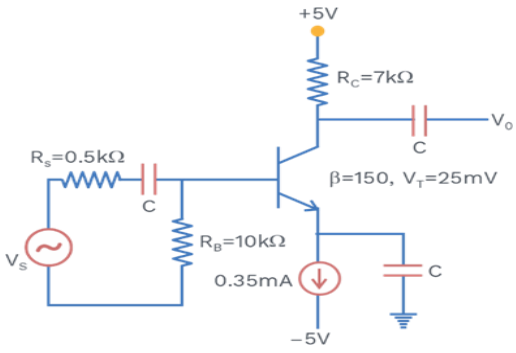
# 4 BJT and MOSFET Amplifier

## MCQ, MSQ and NAT

1. A transistor amplifier circuit is given below. If the transistor $T_1$ has $g_{m_1} = 25.6\ \text{m}\mho, \beta_1 = 59$, and the transistor $T_2$ has $g_{m_2} = 65.3\ \text{m}\mho, \quad \beta_2 = 100$, then the voltage gain $\left(A_v = \frac{V_0}{V_{in}}\right)$ of the amplifier is_____(Rounded off to two decimal places)




2. The voltage gain ($A_{VS}$) of the given circuit is-

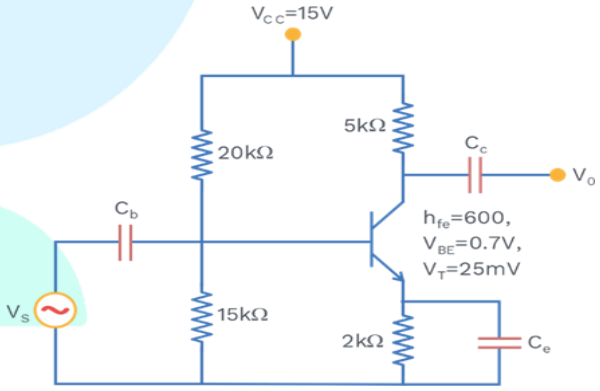



(A)– 10.6 (B)– 1.23

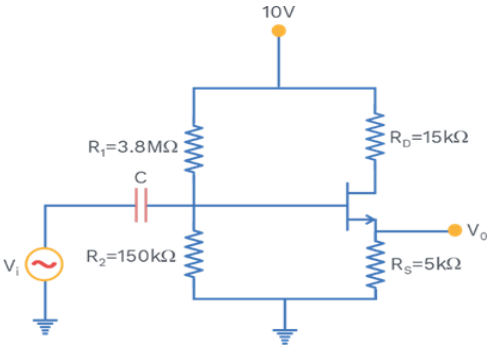
(C)– 104.7 (D)– 89.3

3. A CE-amplifier has $g_m = 15.62\ \text{m}\mho, c_\pi = 4 \times 10^{-6}\ \text{F}$ and $C_\mu = 10^{-7}$ F. The unity gain-bandwidth product of the CE-amplifier is_____Hz. (Up to one decimal place)

4.A self-biased transistor amplifier circuit is shown below. The voltage gain $\left(\left|\frac{V_0}{V_s}\right|\right)$ of the amplifier is_____ . (Up to one decimal place)

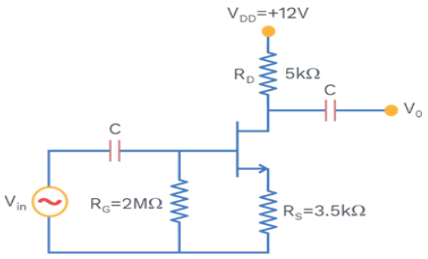



5.The input impedance of the given common drain FET amplifier circuit is _____kΩ. (Up to one decimal place)




6.The common source amplifier

shown below has transconductance $g_m$ = 1.85 mS. The voltage gain of the amplifier is approximately



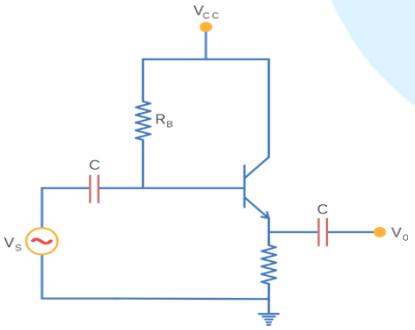

(A)$|A_v|$ = 2.56

(B)$|A_v|$ = 36.5

(C)$|A_v|$ = 1.24

(D)$|A_v|$ = 5.72

7. An amplifier circuit is given below. The given circuit has -



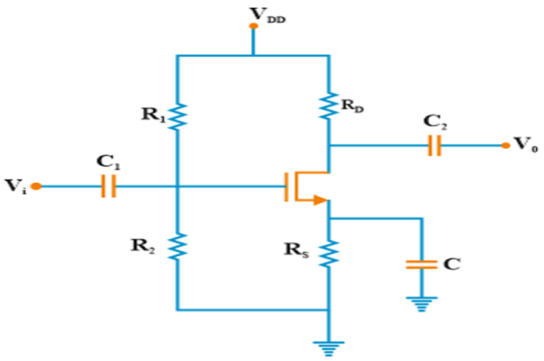

(A)Current sampling series mixing

(B)Voltage sampling series mixing

(C)Voltage sampling shunt mixing

(D)Current sampling shunt mixing

8. A common source amplifier circuit is shown in the figure below.




It has the following parameters:

$K = 0.25\ \text{mA/V}^2$

$V_T = 4\ \text{V}, \quad V_{DD} = 15\ \text{V}, \ g_m = 2000\mu\text{S}$

$R_1 = 60\text{k}\Omega, R_2 = 40\text{k}\Omega, R_D = 6\text{k}\Omega$ and ac input signal is 80mV.

Which of the following statement(s) is/are correct?

(A)Drain current, $I_D$ = 0.5 mA

(B)Drain-source voltage, $V_{DS}$ = 9 V

(C)Voltage Gain, $A_v$ = -12

(D)Output voltage, $V_0$ = -0.96 V

9. A common emitter amplifier circuit has the following parameters:

$h_{ie} = 1000\Omega, h_{fe} = 50, \ h_{re} = 2.5 \times 10^{-4}$

$h_{oe} = 25 \times 10^{-6}\ \text{S}, R_L = 10\text{k}\Omega$ and

$R_S = 100\Omega$

Which of the following statement(s) is/are correct?

(A)Overall current Gain, $A_{is}$ = -4

(B)Overall voltage Gain, $A_{vs}$ = -400

(C)Output conductance, $G_0$ = 13.64S

(D)Output conductance, $G_0 = 13.64\mu\text{S}$

10. The following low-frequency parameters are given for a transistor:

$I_C$ = 5 mA, $V_{CE}$ = 10 V, $h_{fe}$ = 100, $h_{ie}$ = 600 Ω

$|A_i|$ =10 at 10 MHz and $V_T$ = 26 mV

Which of the following statement(s) is/are correct?

(A)$g_m$ = 192.3 mS

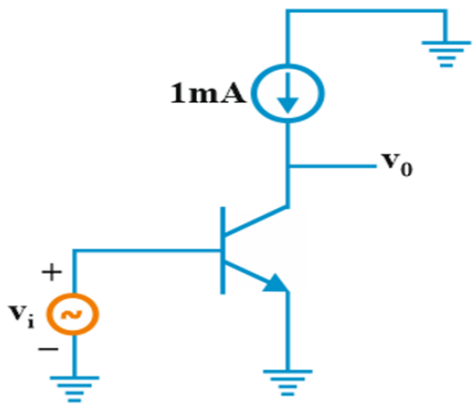
(B)$f_T$ = 100 MHz

(C)$f_\beta$= 1 MHz

(D)$f_\beta$= 10 GHz

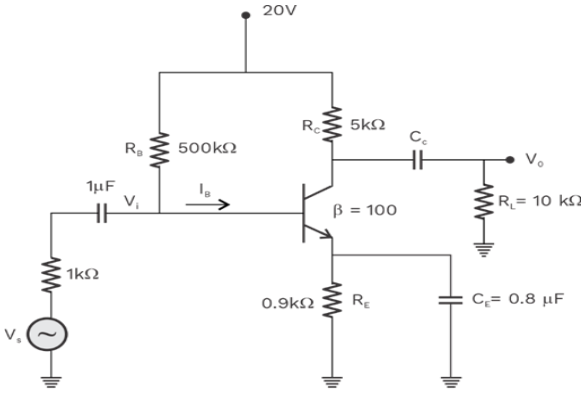
11. Consider that the amplifier circuit shown in the figure below is biased with an ideal current source. If the transistor has β = 100 and early voltage $V_A$ = 10 V then the small signal voltage gain $\left[\frac{V_0}{V_i}\right]$ will be _____. (Rounded off to nearest integer).

[Assume $V_T$ = 25 mV]




12. Consider the circuit shown below.

Assume $V_{BE} = 0.7$ V and $V_T = 26\text{mV}$.



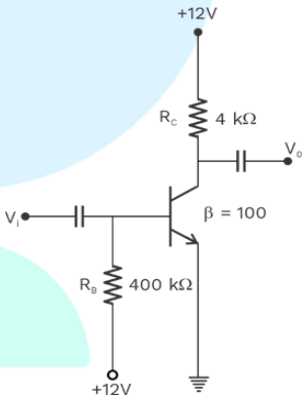

The overall voltage gain will be-

(A)-421.94 (B)-187.08

(C)187.08 (D)421.94

13. For the BJT circuit shown in the figure below,

consider $V_{BE} = 0.7$ V $and V_T = 26\text{mV}$.




Determine the voltage gain (Av).

(A)439

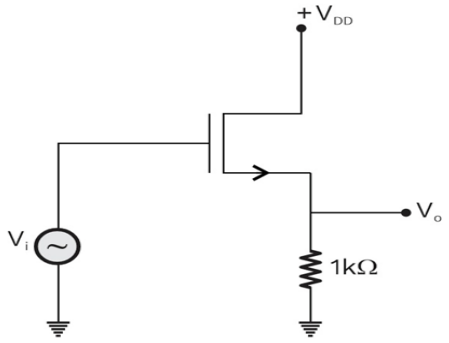
(B)-100

(C)-439

(D)None of the above

14. For the MOSFET circuit given below

consider $r_d = 3k\Omega$ and $\frac{V_0}{V_i} = 0.9.$




The value of transconductance '$g_m$' is _____ mS. (Rounded off to the nearest integer).

15. For the amplifier circuit given below consider-

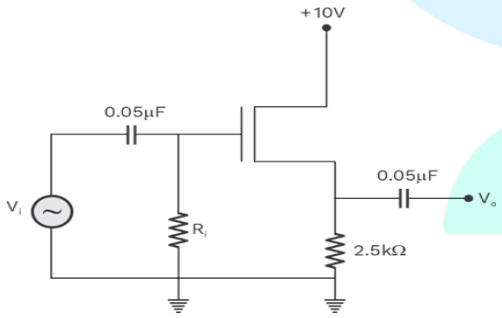
$V_i = 40\text{mV}$

$I_{DSS} = 16\ \text{mA}$

$V_P = -4\ \text{V}$

$V_{GS} = -2.8\ \text{V}$

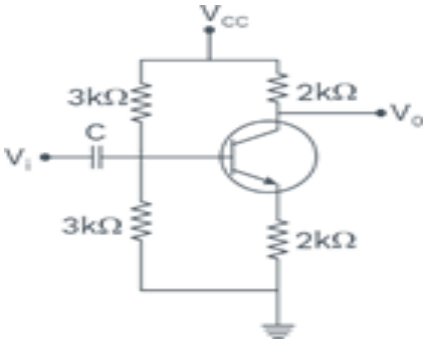
$r_d = 40\text{k}\Omega$




The value of output voltage is_____ mV.[Up to 2 decimal places]

16. A common emitter amplifier is shown below. The quiescent collector voltage of the transistor is

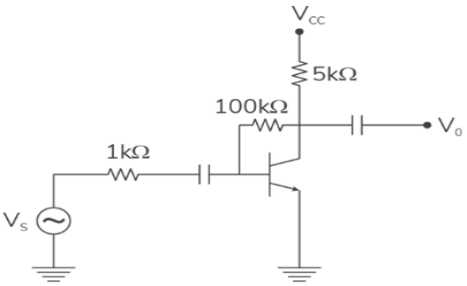
Given, $\beta = 100$ and $\text{V}_{\text{BE}} = 0.7\ \text{V}$




(A)$V_C$ = 8.2 V (B)$V_C$ = 14.8 V

(C) $V_C$ = 6.5 V (D)$V_C$ = 16.2 V

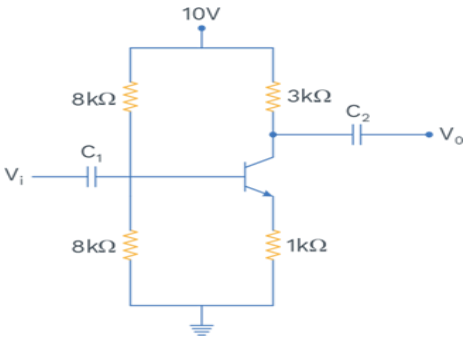
17. For the given BJT circuit with $h_{\text{fe}} = 100$ and $h_{\text{ie}} = 25\text{k}\Omega$. Find the voltage gain $\left(\frac{V_0}{V_S}\right)$.




(A)-100 (B)-57.33

(C)-66.6 (D)-10.5

18. A BJT amplifier is shown in the figure below ( $\beta$ is very high).




Given, $V_T$ = 25mV and $V_{BE}$ = 0.7V. The voltage gain is-

(A)–0.29

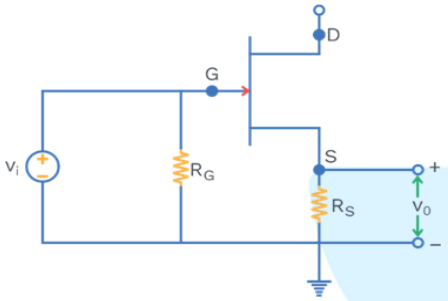
(B)-0.5

(C)-2

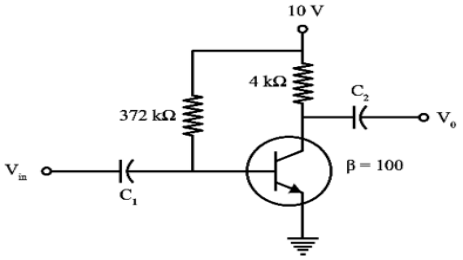
(D)-3

19. In the common drain amplifier shown below, if $R_G$ = 10 MΩ, $R_S$ = 5kΩ, $\mu = 50$ and $r_d$ = 35 kΩ, then the voltage gain $A_V$ of the amplifier is approximately _____.

**(up to two decimal places).**




20. Consider an amplifier circuit shown in the figure below:




The magnitude of voltage gain of the circuit is equal to

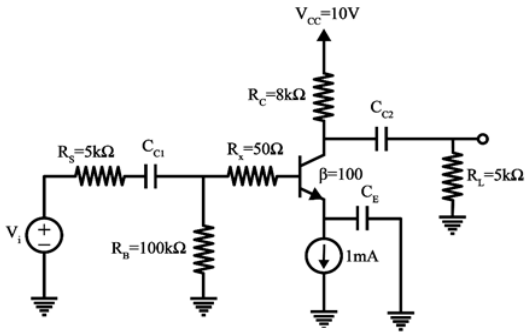
$(Assume V_t = 25mV$ and $V_{BE} = 0.7\ V$

(A)415 (B)200

(C)400 (D)210

21. For the common emitter amplifier shown below, if the unity gain frequency is 800MHz

$C_\mu = 1pF, C_\pi = 7pF, r_\pi = 2.5k\Omega, r_0 = 100k\Omega$, and the upper 3dB cut-off frequency $f_H$ (in kHz) and mid band gain are



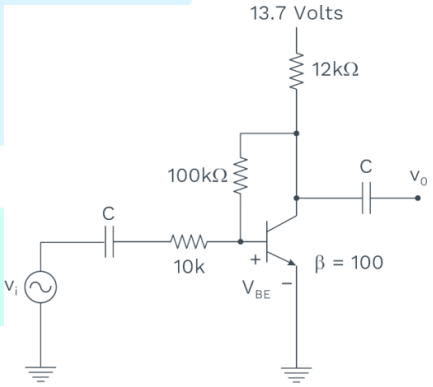

(A)A = 120.16

(B)$f_H = 762.1kHz$

(C)A = -119.4

(D)$f_H = 280.2kHz$

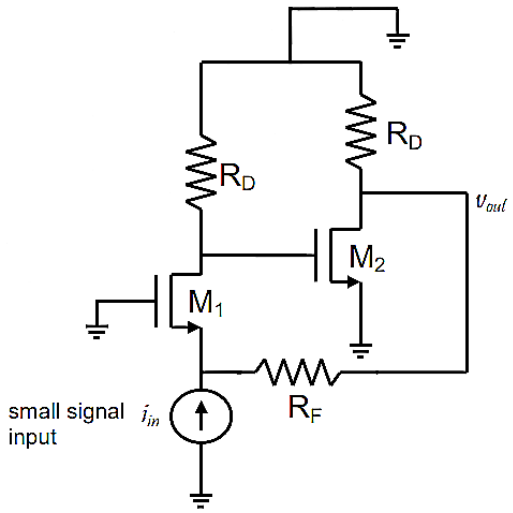
22. The voltage gain $A_v$ of the circuit shown below is




(A) $|A_v| \gg 200$ (B) $|A_v| \approx 100$

(C) $|A_v| \approx 20$ (D) $|A_v| \approx 10$

23. In the ac equivalent circuit shown in the figure, if $i_{in}$ is the input current and $R_F$ is very large, the type of feedback is



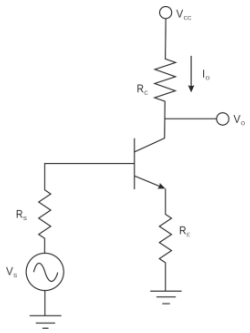

(A) voltage-voltage feedback

(B) voltage-current feedback

(C) current-voltage feedback

(D) current-current feedback

24. The feedback topology in the amplifier circuit (the base bias circuit is not shown for simplicity) in the figure is



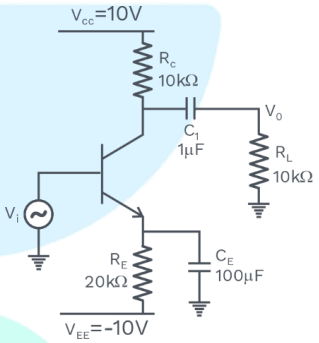
(A) Voltage shunt feedback

(B) Current series feedback

(C) Current shunt feedback

(D) Voltage series feedback

25. For the BJT in the amplifier shown below, $V_{BE}$=0.7V,kT/q=26mV. Assume that BJT output resistance ($r_0$) is very high and the base current is negligible. The capacitors are also assumed to be short circuited at signal frequencies. The input $V_i$ is direct coupled. The low frequency voltage gain $V_0/V_i$ of the amplifier is:




(A) –89.42 (B) –128.21

(C) –178.85 (D) –256.42

■■■■

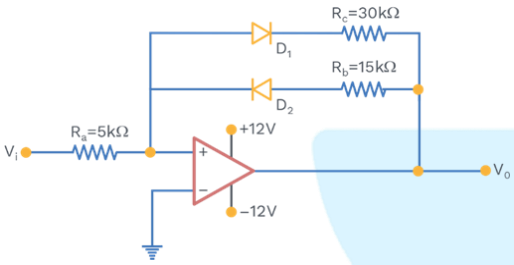
## Answer Key

| | | |
|---|---|---|
| Q.1 | Ans. | [-2.51] |
| Q.2 | Ans. | [ D ] |
| Q.3 | Ans. | [606.3] |
| Q.4 | Ans. | [ 572.5 ] |
| Q.5 | Ans. | [144.3] |
| Q.6 | Ans. | [C] |
| Q.7 | Ans. | [C] |
| Q.8 | Ans. | [B ,C ,D ] |
| Q.9 | Ans. | [A ,B ,D] |
| Q.10 | Ans. | [ A ,B ,C ] |
| Q.11 | Ans. | [-16] |
| Q.12 | Ans. | [B] |
| Q.13 | Ans. | [ C ] |
| Q.14 | Ans. | [12] |
| Q.15 | Ans. | [33.96] |
| Q.16 | Ans. | [ B ] |
| Q.17 | Ans. | [ B ] |
| Q.28 | Ans. | [D] |
| Q.19 | Ans. | [0.86] |
| Q.20 | Ans. | [ C ] |
| Q.21 | Ans. | [B ,C] |
| Q.22 | Ans. | [D] |
| Q.23 | Ans. | [B] |
| Q.24 | Ans. | [B] |
| Q.25 | Ans. | [A] |

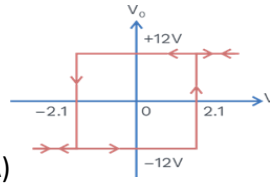
# 5 Feedback Amplifier and Oscillator

## MCQ, MSQ and NAT

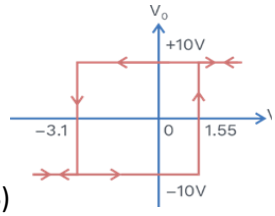
1. For the circuit shown in the figure below, the output saturates at ± 10 V, and the forward voltage drop of the diodes is 0.7 V. The transfer characteristics is -



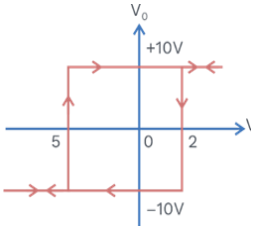

(A)



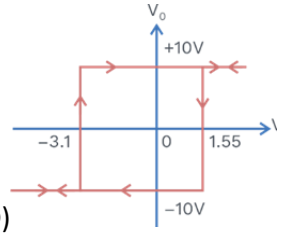

(B)



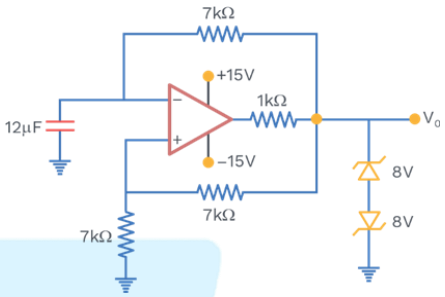

(C)




(D)




2. A circuit is shown in shown on the figure below.




The supply voltage is ±15V. The output waveform has-

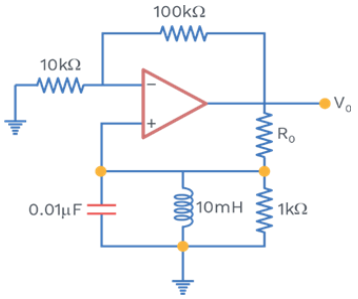
(A)an amplitude of ± 15V and a time period of 58.22 msec.

(B)an amplitude of ± 8V and a time period of 116.44 msec.

(C)an amplitude of ± 8V and a time period of 58.22 msec.

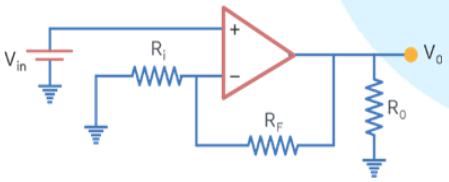
(D)an amplitude of ± 15V and a time period of 116.44 msec.

3. A circuit is shown in the figure below. The value of 'R' at which the circuit oscillates is _____ kΩ. (Rounded off to the nearest integer).




4. A circuit consisting of two identical amplifiers having a voltage gain of 75 each are cascaded. The input and output resistances are 2 kΩ and 400 Ω respectively. The open-circuit voltage gain of the cascaded amplifier is _____dB. (Up to 2 decimal places)

5. A circuit is shown in the figure below.




The feedback topology is-

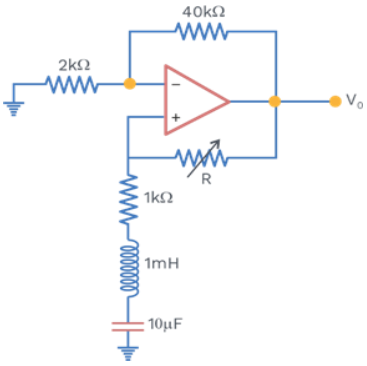
(A)Current series

(B)Voltage-series

(C)Current shunt

(D)Voltage-shunt

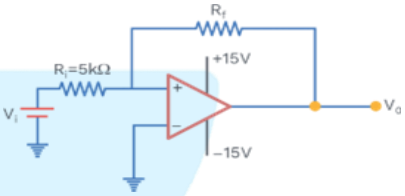
6. A sinusoidal oscillator circuit is shown below. The value of 'R' for sinusoidal oscillations is-




(A)$20\ \Omega$ (B)$10\ k\Omega$

(C)20 kΩ (D)0 Ω

7. A circuit is shown in the figure below.




Assume that the width of hysteresis is equal to 250 mV. The value of $R_f$ is _____ $k\Omega$. (Round off to the nearest integer)

8.Two amplifiers having the lower cut-off frequency of 5 MHz each are cascaded. The lower cut-off frequency of the cascaded system is_____ MHz. (Up to two decimal places)

9. A tuned amplifier circuit with capacitive coupling is having $R = 10\Omega, L = 20\text{mH}$, and $C = 0.05\mu\text{F}$. The bandwidth of the amplifier is _____ Hz. (Up to two decimal places)

10. An amplifier has an input of 10 mV and a gain of 200(without feedback). The distortion produced at the output of the

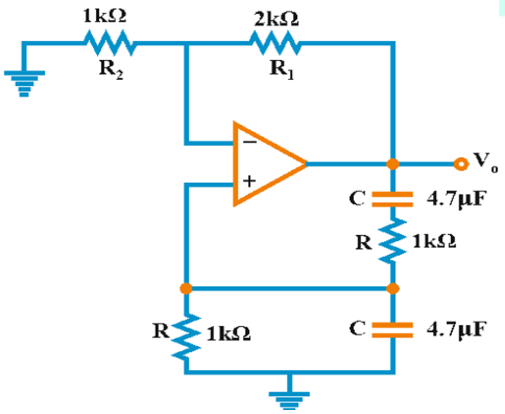
amplifier is 10%. It is required to reduce the distortion to 1% by using negative feedback. The output voltage with applied feedback is _____V. (Up to one decimal place)

11. In a transistor-based Hartley oscillator, the tank circuit has a capacitance of 100pF. The value of inductance between the collector and the tapping point is 30mH and the value of inductance between the tapping point and transistor base is 100$\mu H$ . Assume the mutual inductance is $50\mu H$. The frequency of oscillation of the oscillator is-

(A)91.73 Hz (B)81.58 Hz

(C)91.58 kHz (D)81.73 kHz

12. Find the frequency of oscillation for the given circuit. Assume op-amp to be ideal.




(A)33.86 rad/sec

(B)33.86 Hz

(C)212.76 Hz

(D)212.76 kHz

13. A voltage–series feedback amplifier is having an open loop gain $A_v$ = 300, input resistance $R_i = 1.5k\Omega$, output resistance $R_0 = 50k\Omega$, and feedback factor $\beta = \frac{1}{15}$.

Which of the following statement(s) is/are correct?

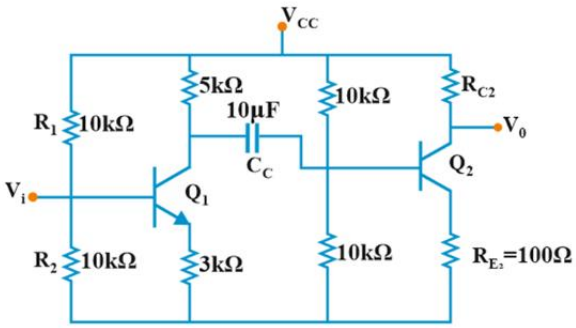
(A)With feedback, gain $A_{vf}$ = 14.28

(B)With feedback, gain $A_{vf}$ = 6300

(C)Input resistance with feedback $R_{if} = 31.5k\Omega$

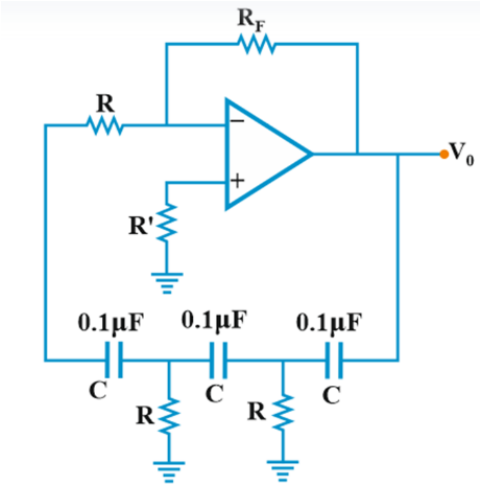
(D)Output resistance with feedback, $R_{of} = 2.38k\Omega$

14. A multistage amplifier circuit is shown in the figure below.




The transistors $Q_1$& $Q_2$ are biased such that the transconductance of both the transistors is 20 mA/V. If β of both the transistors is 100, then the 3-dB cut-off frequency due to coupling capacitor $C_c$ is _____ Hz. (Up to two decimal places)

15. Consider the given RC phase shift oscillator.




Op-amp is assumed to be ideal. The minimum value of $R_F$ which will sustain oscillations at a frequency of 100 Hz is _____ kΩ. (Up to two decimal places)

16. A single-stage basic amplifier with open loop gain $A_V$ = 1000 has negative feedback of $\beta = \frac{1}{10}$. The lower cut-off frequency $f_L$= 20 Hz and the upper cut-off frequency $f_H$ = 40 kHz.

Which of the following is/are correct?

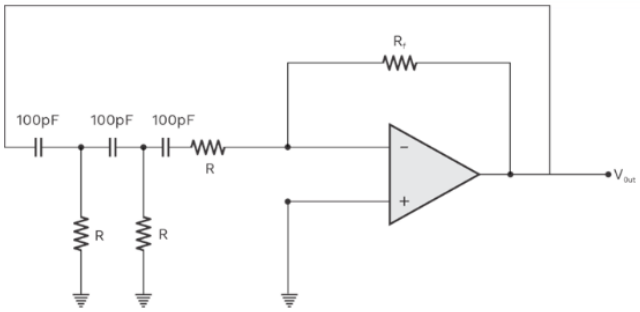
(A)Lower cut-off frequency with feedback, $f'_L = 0.198\text{kHz}$

(B)Lower cut-off frequency with feedback, $f'_L = 0.198\ \text{Hz}$

(C)Higher cut-off frequency with feedback, $f'_H = 4.04\text{MHz}$

(D)Higher cut-off frequency with feedback, $f'_H = 4.04\text{kHz}$

17. Consider the circuit shown below.



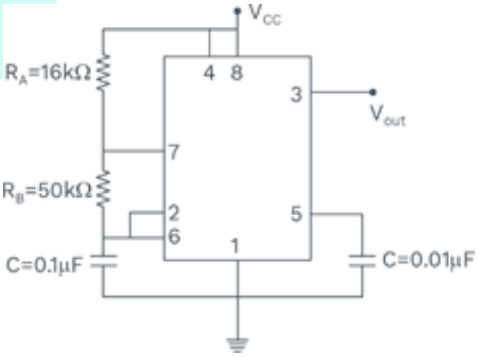

If the frequency of oscillations of the above circuit is 200 kHz, then the value of R will be_____ k Ω. [up to 2 decimal places]

18. An amplifier has $A_v = 100, f_L = 5\text{kHz}, f_H = 250\text{kHz}$, and negative feedback with $\beta = 0.1$. The Bandwidth of the feedback amplifier is-

(A)403.2 kHz (B)3049.4 kHz

(C)505.9 kHz (D)2749.5 kHz

19. The circuit of 555 timer working as an astable multivibrator is shown below




The duty cycle of the output voltage is:

(A)33.3% (B)75%

(C)55.6% (D)60%

20. When we cascade 'n' number of amplifiers, then

(A)$f_H$ decreases, $f_L$ increases

(B)$f_H$ decreases, $f_L$ decreases

(C)$f_H$ increases, $f_L$ increases

(D)$f_H$ increases, $f_L$ decreases

21. When two identical stages with upper cut off at $\omega_n$ are cascaded, the overall upper cut-off point is:

(A)$\omega_n$ (B)2 $\omega_n$

(C)0.5 $\omega_n$ (D) 0.64 $\omega_n$

22. An amplifier with an initial open-loop gain of 800 is used as a negative feedback amplifier. The feedback factor is 0.1. If the gain of the amplifier changes by 20% due to temperature, then the closed-loop gain will change approximately by

(A)0.25% (B)0.1%

(C)5% (D)0.5%

23. A Colpitts oscillator has a coil with an inductance of 25 µH and is tuned by a capacitor of 200 pF across the amplifier input and 50 pF across the output. The frequency of oscillation and minimum gain for maintaining oscillation are:

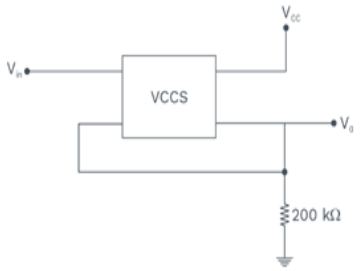
(A)2.5 MHz, 10

(B)10 MHz, 4

(C)5 MHz, 10

(D)5 MHz, 4

24. For the Voltage Controlled Current Source(VCCS) circuit shown below, the approximate output impedance at $V_0$ is –




[Consider $g_m = 0.02S \; and \; R_{in} = 20M\Omega$]

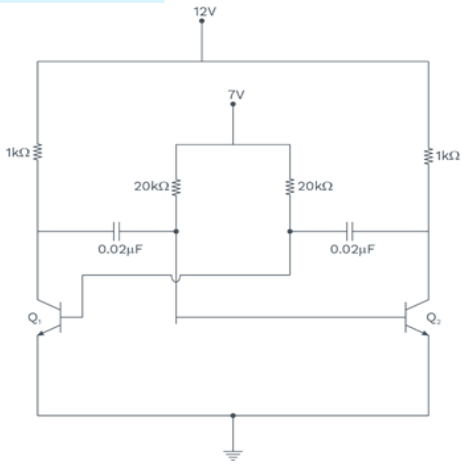
(A) $10\Omega$

(B)$50\Omega$

(C)$100\Omega$

(D)None of the above

25. For the circuit shown below, the frequency of oscillation is-




(A)5.2 kHz (B)2.5 kHz

(C)2 kHz (D)1.8 kHz

26. Two amplifiers of upper cutoff frequency 2MHz are cascaded, the upper cutoff frequency of the cascaded system is

(A)4 MHz

(B)2 MHz

(C)1.28 MHz

(D)2.56 MHz

27. Which of the following converts

irregularly shaped waveform into a regular-shaped waveform?

(A)Schmitt trigger

(B)Comparator

(C)Voltage limiter

(D)MOSFET

28. Which circuit is used as amplitude comparator?

(A) Bistable

(B)Monostable

(C)Astable

(D)Schmitt trigger

29. The frequency of oscillations of transistorized Colpitts oscillator having tank circuit parameters C1 = 150 pF, C2 = 1.5 nF and L = 50 μH is-

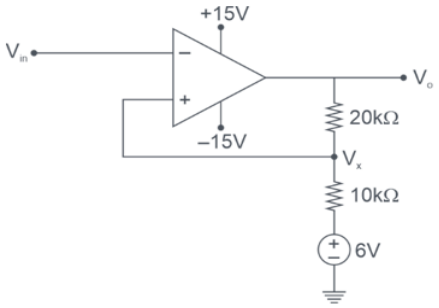
(A)500 kHz

(B)3.26 MHz

(C)1.927 MHz

(D)63.5 kHz

30. Operation amplifier circuit shown below, the upper and lower threshold voltage for the circuit is respectively



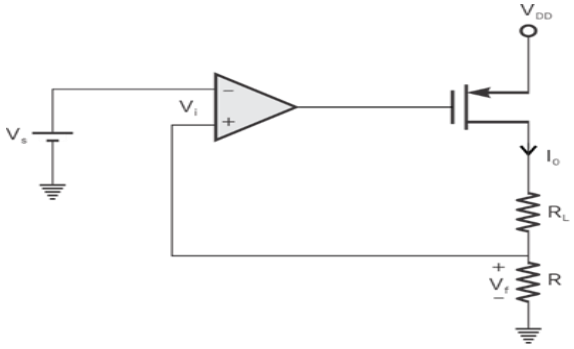

(A)+1V and –9V

(B)+1V and –1V

(C)+9V and –9V

(D)+9V and –1V

31. The type of feedback topology for the circuit shown below is-



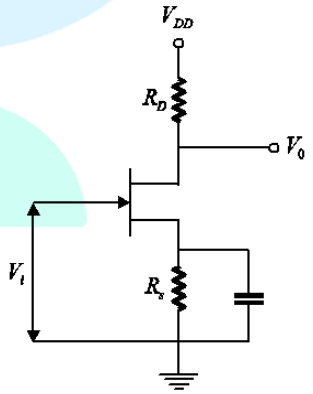

(A)Voltage mixing and voltage sampling

(B)Current mixing and voltage sampling

(C)Voltage mixing and current sampling

(D)Current mixing and current sampling

32. Feedback topology in the circuit shown in figure is



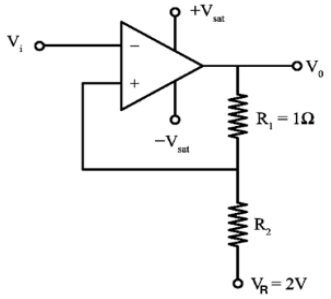

(A)No feedback

(B)Current series

(C)Voltage series

(D)None

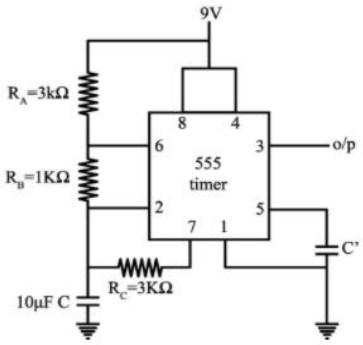
33. The maximum output voltage of given circuit is ± 10V, the hysteresis voltage is 0.4V. Then, the value of $R_2$ (in ohms) would be




(A)$49\Omega$ (B)$\frac{1}{49}\Omega$

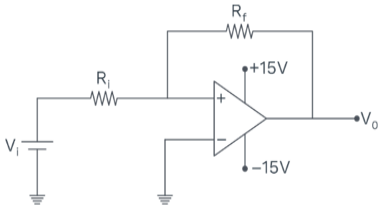
(C)$98\Omega$ (D)$\frac{1}{98}\Omega$

34. The Duty cycle of the output in (%) for below Astable multivibrator is_____.




35. An amplifier without feedback has a voltage gain of 60, input resistance of 1.5kΩ and output resistance of 3kΩ. If the feedback factor of the above amplifier is 0.3, then the input resistance of the current shunt negative feedback amplifier is _____ kΩ (up to 2 decimal places)

36. A circuit is shown in the figure below.

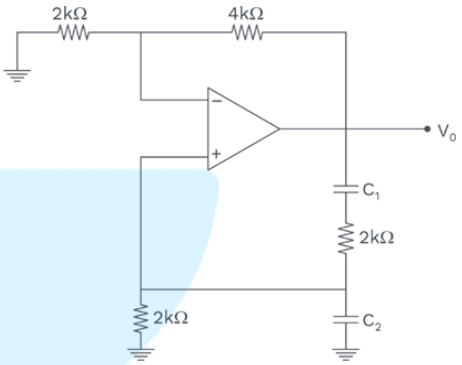



The circuit $R_i = 2k\Omega$ and hysteresis of $200\text{mV}$ width. The value of $R_f$ is -

(A)200 kΩ (B)300 kΩ

(C)450 kΩ (D)500 kΩ

37. For the circuit shown in the figure below, calculate the value of the capacitor which is required for sinusoidal oscillation of frequency of 1 kHz in the circuit. (Assume that the capacitors are identical).




(A)$4\pi\sqrt{6}\mu\text{F}$ (B)$\frac{1}{4\pi\sqrt{6}}\mu\text{F}$

(C)$4\pi\mu\text{F}$ (D)$\frac{1}{4\pi}\mu\text{F}$

38. To obtain a very high input and output impedance in a feedback amplifier, the feedback mostly used is -

(A)current series

(B)voltage series

(C)current shunt

(D)voltage shunt

39. The figure (a) shows a Schmitt trigger circuit and figure (b) the corresponding hysteresis characteristics. The values of $V_{TL}\&V_{TH}$ are

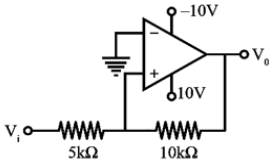



Fig: a

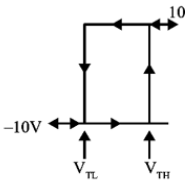



Fig: b

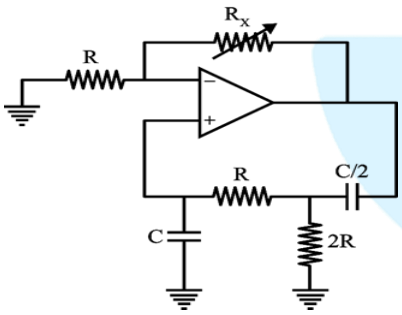
(A)$\mathrm{V_{TL}} = -5\ \mathrm{V}$

(B)$V_{TH} = 1V$

(C)$\mathrm{V_{TL}} = -3.57\ \mathrm{V}$

(D)$\mathrm{V_{TH}} = 5\ \mathrm{V}$

40. The ratio of $\mathrm{R_x/R}$ to have sustained oscillations in the given oscillator circuit is _____.




41. Two amplifiers of the upper cutoff frequency of 2MHz are cascaded, the upper cutoff frequency of the cascaded system is _____.

(A)4 MHz (B)2 MHz

(C)1.28 MHz (D)2.56 MHz

42. The frequency of oscillations of transistorized Colpitts oscillator having tank circuit parameters,

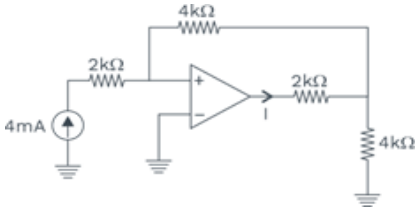
$\mathrm{C_1 = 160pF}$ , $\mathrm{C_2 = 1.6nF}$ and $\mathrm{L = 50}$ $\mu\mathrm{H}$ is-

(A)3 MHz (B)200 kHz

(C)1.87 MHz (D)50 kHz

43. For the circuit shown below., find the value of current I.

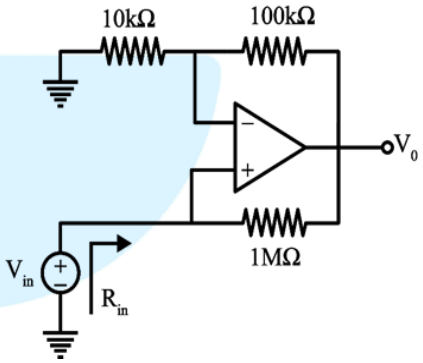
Assume that the op-amp is ideal.




(A)4 mA (B) –8mA

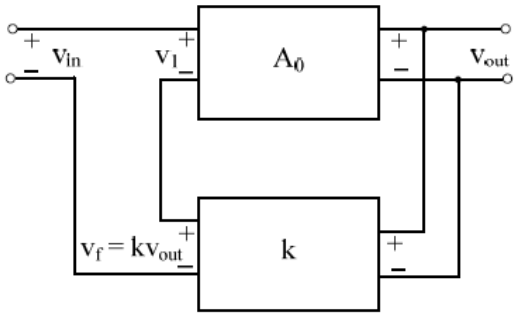
(C)8mA (D)–2mA

44. Consider the op-amp circuit shown in the figure below:




The resistance $R_{in}$as seen by the voltage source is _____kΩ.

45. In a voltage-voltage feedback as shown below, which one of the following statements is TRUE if the gain $k$ is increased?




(A) The input impedance increases and output impedance decreases.

(B) The input impedance increases and output impedance also increases.

(C) The input impedance decreases and output impedance also decreases.

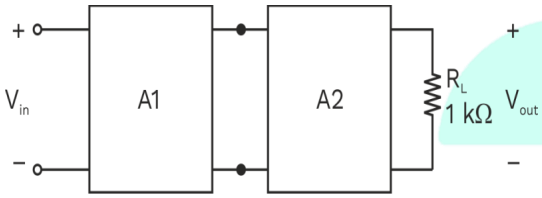
(D) The input impedance decreases and output impedance increases.

46. A cascade connection of two voltage amplifiers A1 and A2 is shown in the figure. The open-loop gain $A_{v0}$, input resistance $R_{in}$, and output resistance $R_o$ for A1 and A2 are as follows:

A1: $A_{v0} = 10, R_{in} = 10k\Omega, R_o = 1k\Omega$

A2: $A_{v0} = 5, R_{in} = 5k\Omega, R_o = 200\,\Omega$

The approximate overall voltage gain $v_{out} / v_m$ is _____.




Q.47 The desirable characteristics of a transconductance amplifier are

A. high input resistance and high output resistance

B. high input resistance and low output resistance

C. low input resistance and high output resistance

D. low input resistance and low output resistance

## Answer Key

| | | |
|---|---|---|
| Q.1 | Ans. | [ B ] |
| Q.2 | Ans. | [ B ] |
| Q.3 | Ans. | [ 10 ] |
| Q.4 | Ans. | [73.41] |
| Q.5 | Ans. | [B] |
| Q.6 | Ans. | [C] |
| Q.7 | Ans. | [600] |
| Q.8 | Ans. | [7.78] |
| Q.9 | Ans. | [79.59] |
| Q.10 | Ans. | [0.2] |
| Q.11 | Ans. | [C] |
| Q.12 | Ans. | [B] |
| Q.13 | Ans. | [A ,C ,D] |
| Q.14 | Ans. | [1.81] |
| Q.15 | Ans. | [188.21] |
| Q.16 | Ans. | [B ,C] |
| Q.17 | Ans. | [3.25 |
| Q.18 | Ans. | [ D ] |
| Q.19 | Ans. | [D] |
| Q.20 | Ans. | [A] |
| Q.21 | Ans. | [D] |
| Q.22 | Ans. | [A] |
| Q.23 | Ans. | [D] |
| Q.24 | Ans. | [B] |
| Q.25 | Ans. | [D] |
| Q.26 | Ans. | [C] |
| Q.27 | Ans. | [A] |
| Q.28 | Ans. | [D] |
| Q.29 | Ans. | [C] |
| Q.30 | Ans. | [D] |
| Q.31 | Ans. | [C] |
| Q.32 | Ans. | [B] |
| Q.33 | Ans. | [ B ] |
| Q.34 | Ans. | [57.14] |
| Q.35 | Ans. | [0.15] |
| Q.36 | Ans. | [ B ] |
| Q.37 | Ans. | [ D ] |
| Q.38 | Ans. | [ A ] |
| Q.39 | Ans. | [ A ,D ] |
| Q.40 | Ans. | [3] |
| Q.41 | Ans. | [C] |
| Q.42 | Ans. | [C] |
| Q.43 | Ans. | [B] |
| Q.44 | Ans. | [-100] |
| Q.45 | Ans. | [A] |
| Q.46 | Ans. | [34.722] |
| Q.47 | Ans. | [A] |